

Human-Computer Interaction

Johan F. HOORN

Lecture 03: Visual perception and computer graphics, Part 1

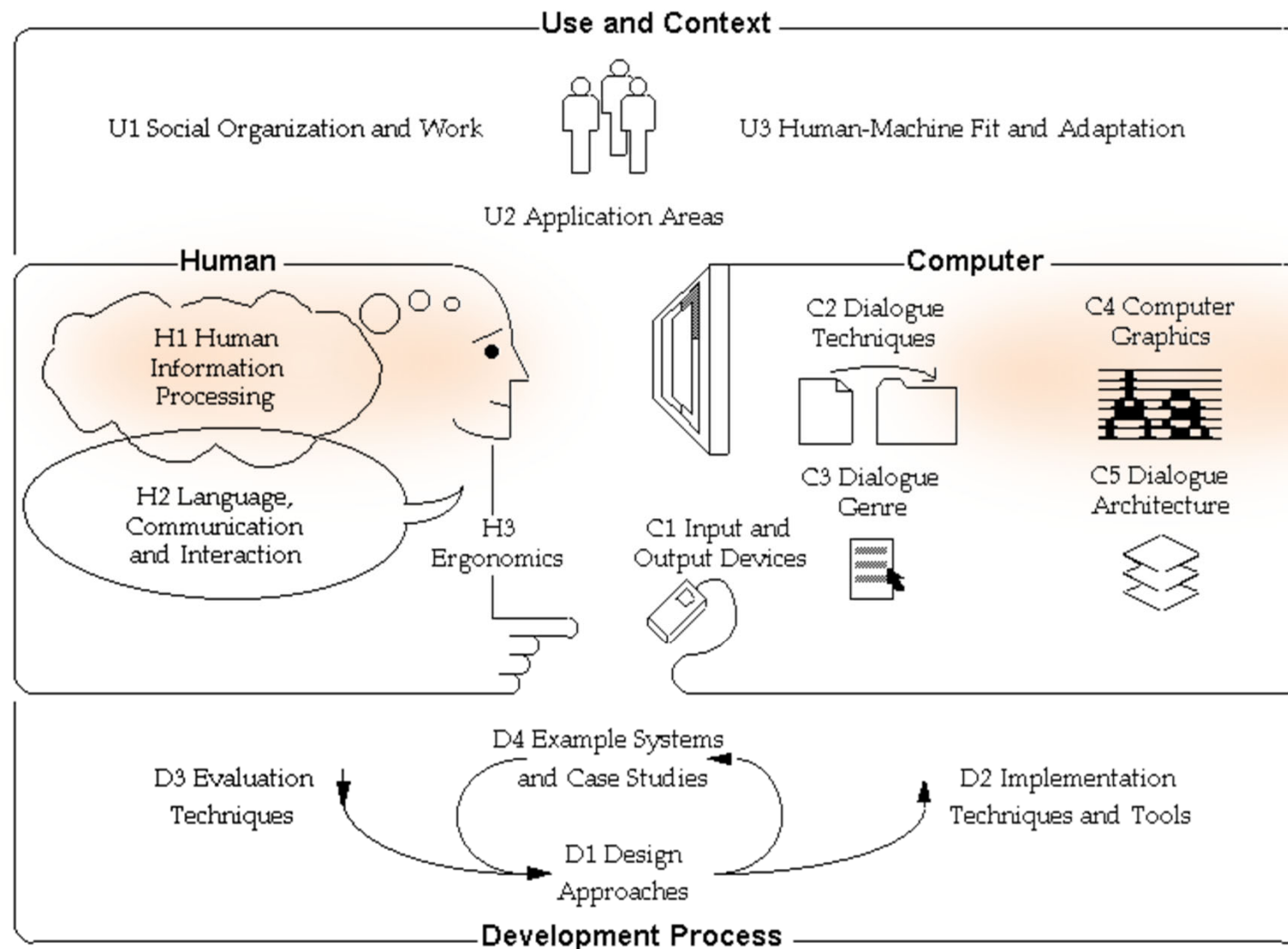


“I am programmed to understand humans”

C-3PO (“3PO” aka Threepio)

Lukas, G., & Hales, J. (2002). *Star Wars Episode II: Attack of the Clones*. Los Angeles, CA: 20th Century Fox. [Part 7/10] - 1080p HD, 7:00 min. Retrieved May 21, 2012 from http://www.youtube.com/watch?v=-o5Y_AszGg

Overview



Contents

☐ Visual

- Image
 - Physical reception
 - Interpretation: 3D, colour
 - Images in design
- Text
 - Physical reception
 - Interpretation: semantics, pragmatics
 - Text in design
- Computer Graphics

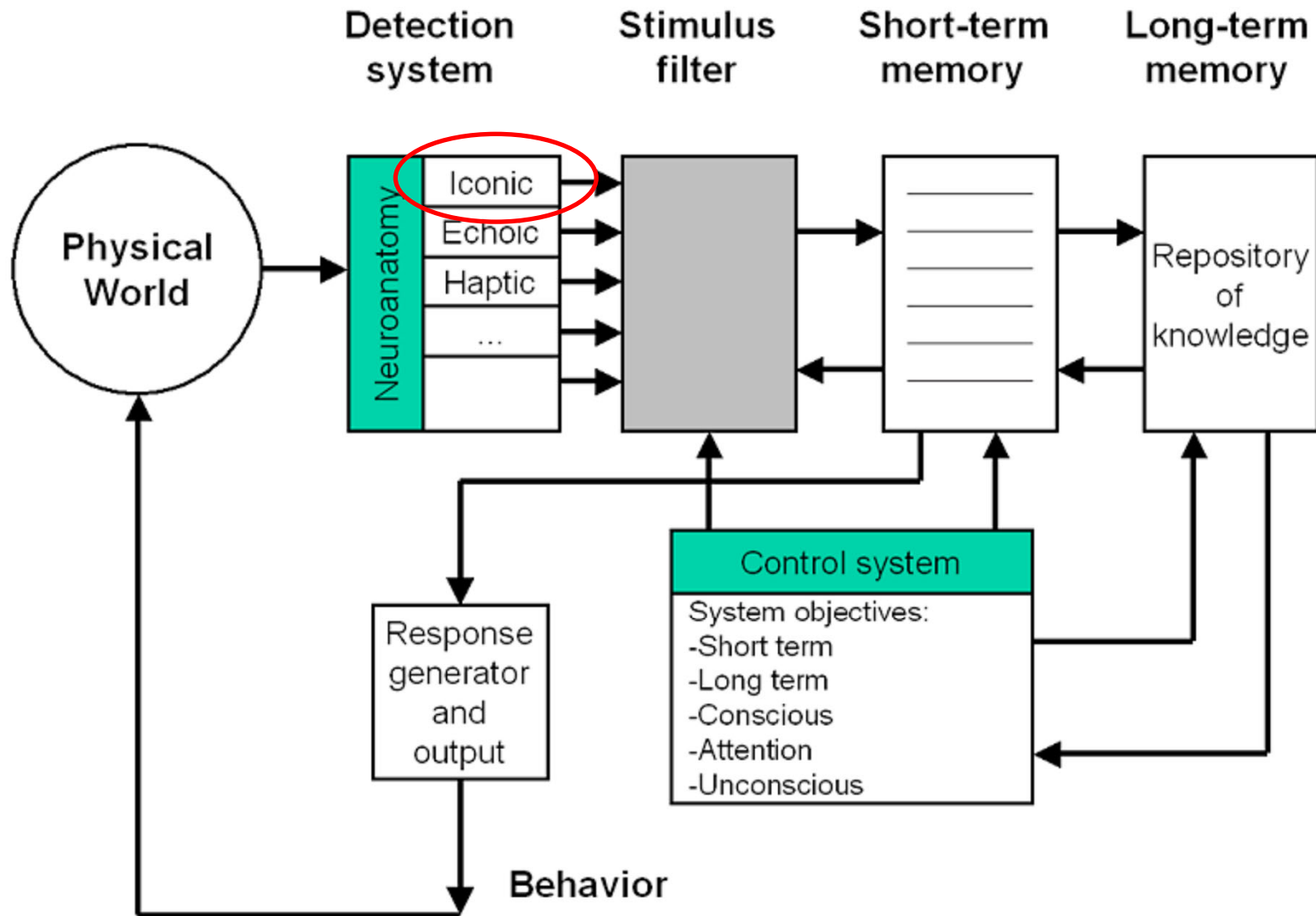


What you'll learn today

- ☐ Human visual system
- ☐ Perceiving images
- ☐ Detecting patterns ('Gestalts')
- ☐ Deciphering texts
- ☐ How subjective and deceptive all this is
- ☐ Consequences for design



The Human Information Processor

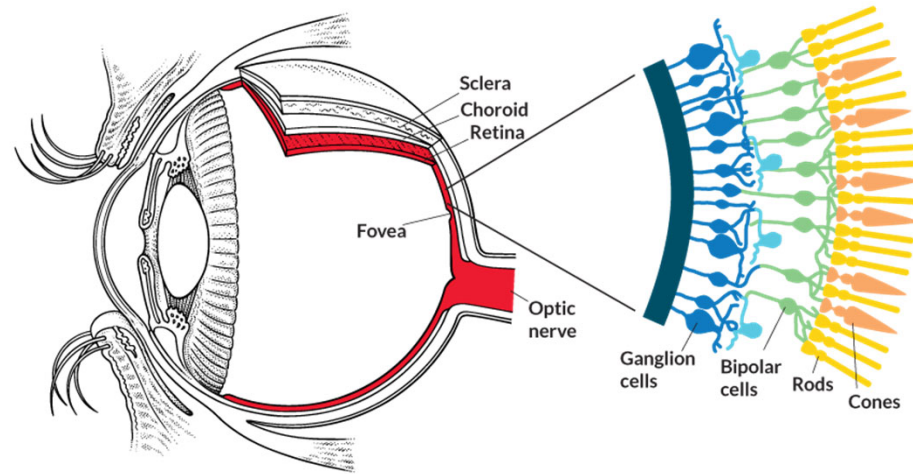


Visual system

Input: perception

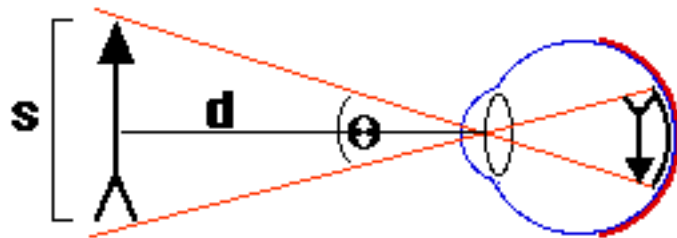
□ Visual ('iconic')

- Image
 - Physical reception



Eye receives light and transforms it into electrical energy

Light reflects from objects; their images are focused upside-down on retina



Law of visual angle

$$\tan \theta/2 = s/2d$$

$$\theta = 2 \arctan(s/2d)$$

$$\theta \quad s = 2d \tan \theta/2$$

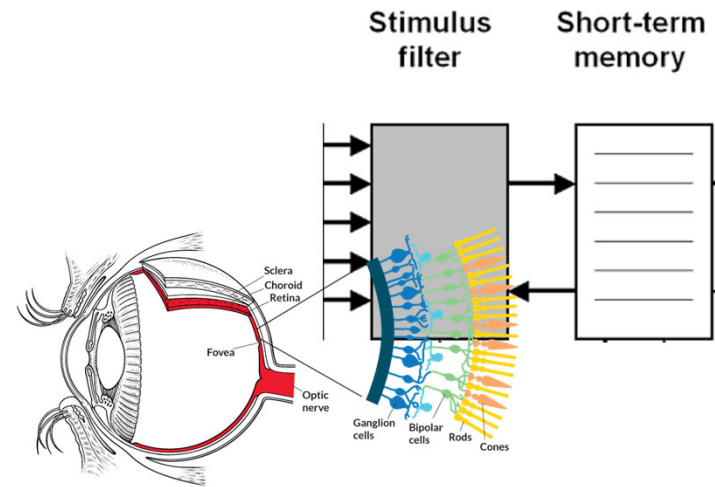
Ganglion cells detect pattern and movement

Retina contains rods for low-light vision and cones for colour vision

Input: perception

☐ Visual ('iconic')

- Image
 - Aspects of physical reception



Brightness

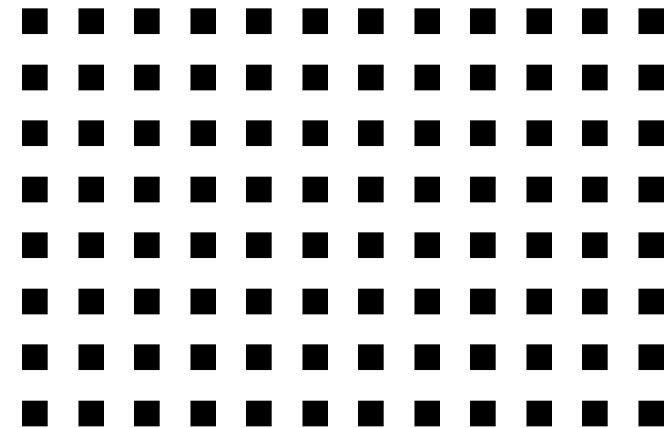
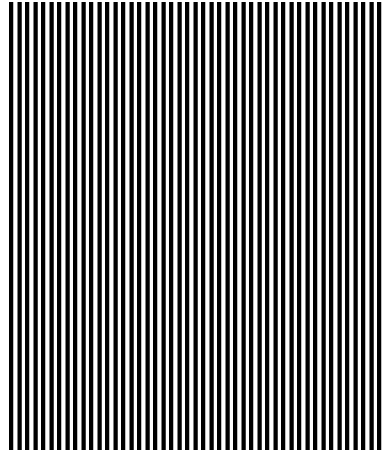
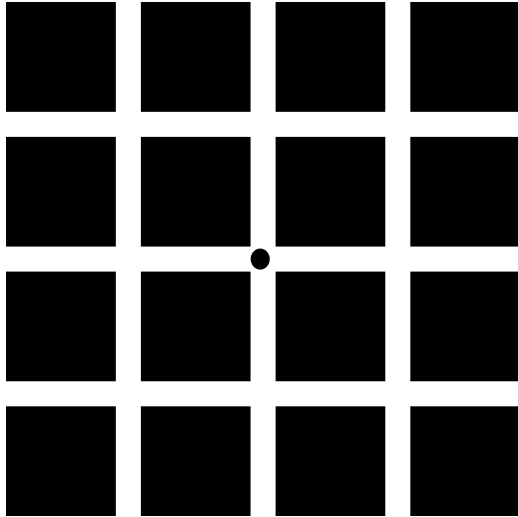
- subjective reaction to levels of light
- affected by luminance of object
- measured by just noticeable difference
- visual acuity increases with luminance as does flashing



Pattern filters (ganglion cells)

- certain neurons work as pattern filters
- filters pass middle frequencies better than higher and lower frequencies
- mutual suppression
(random noise from individual cells is corrected by the larger group)

Hermann grid



Springer lines

Are the grey spots and lines visual input or output?

Which colour reflects the higher frequency?

Which the lower?

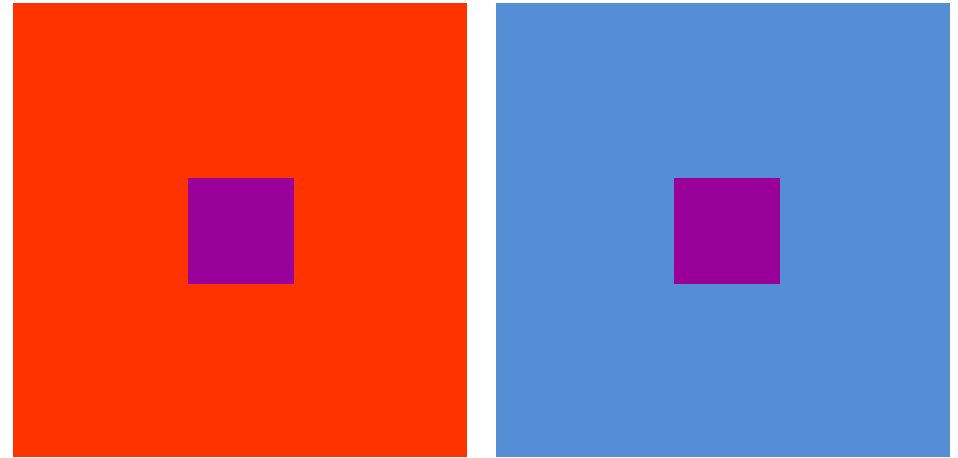
What the middle?

Is vision completely data-driven?

Input: perception

☐ Visual

- Image
 - Physical reception



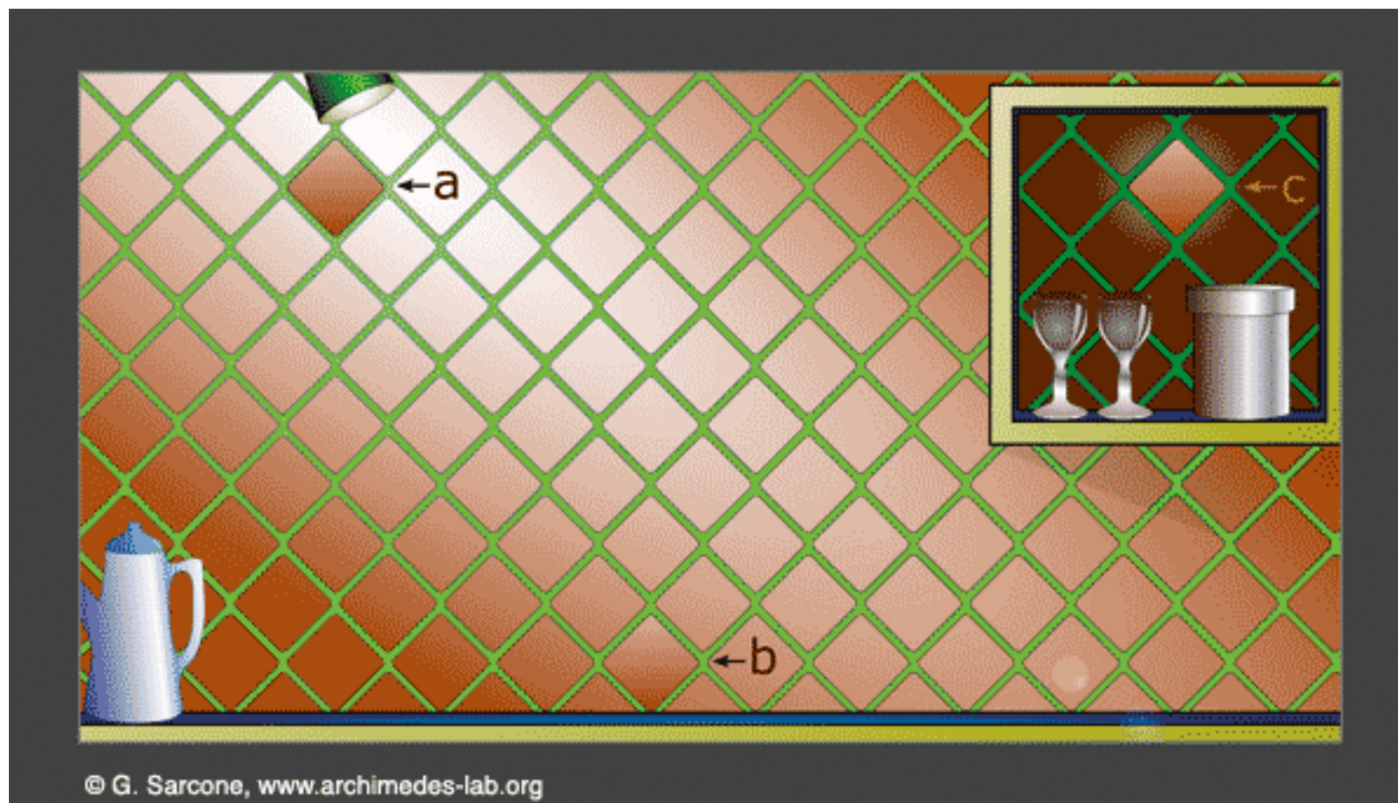
Colour

- Hue (based on the wavelength of light)
- Intensity (lighter/darker)
- Saturation (purity of the hue in a colour)
- The higher the saturation, the purer the colour
- Cones sensitive to colour wavelengths
- Again, mutual suppression
- Blue acuity is lowest
- 8% males and 1% females are colour blind

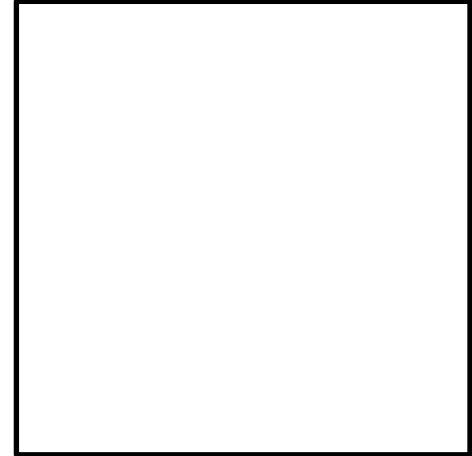
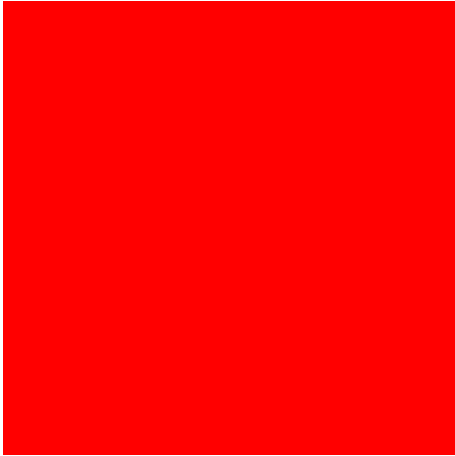
Input: perception

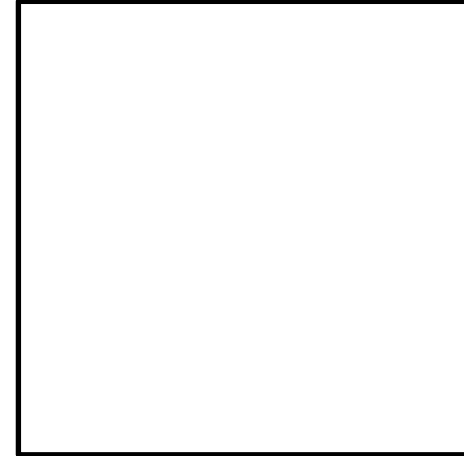
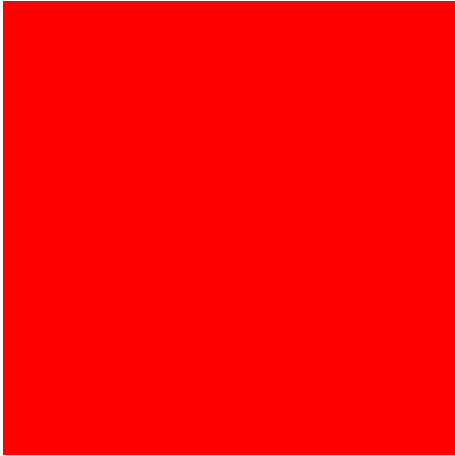
□ Visual

- Image
 - Physical reception

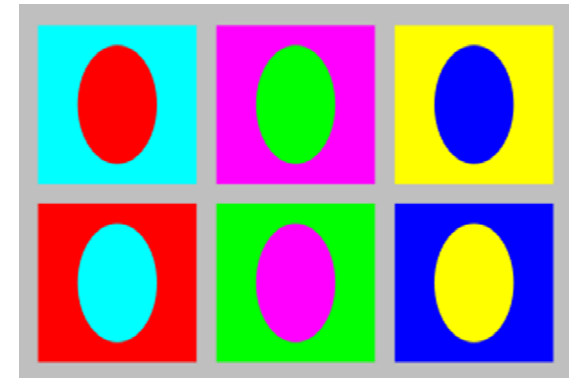


Which square is the darkest?





- Cones for red wear out
- Mutual suppression (e.g., red over green)
- Cones for cyan remain firing
- After-images



Complementary colours
in the [RGB color model](#)

Input: perception

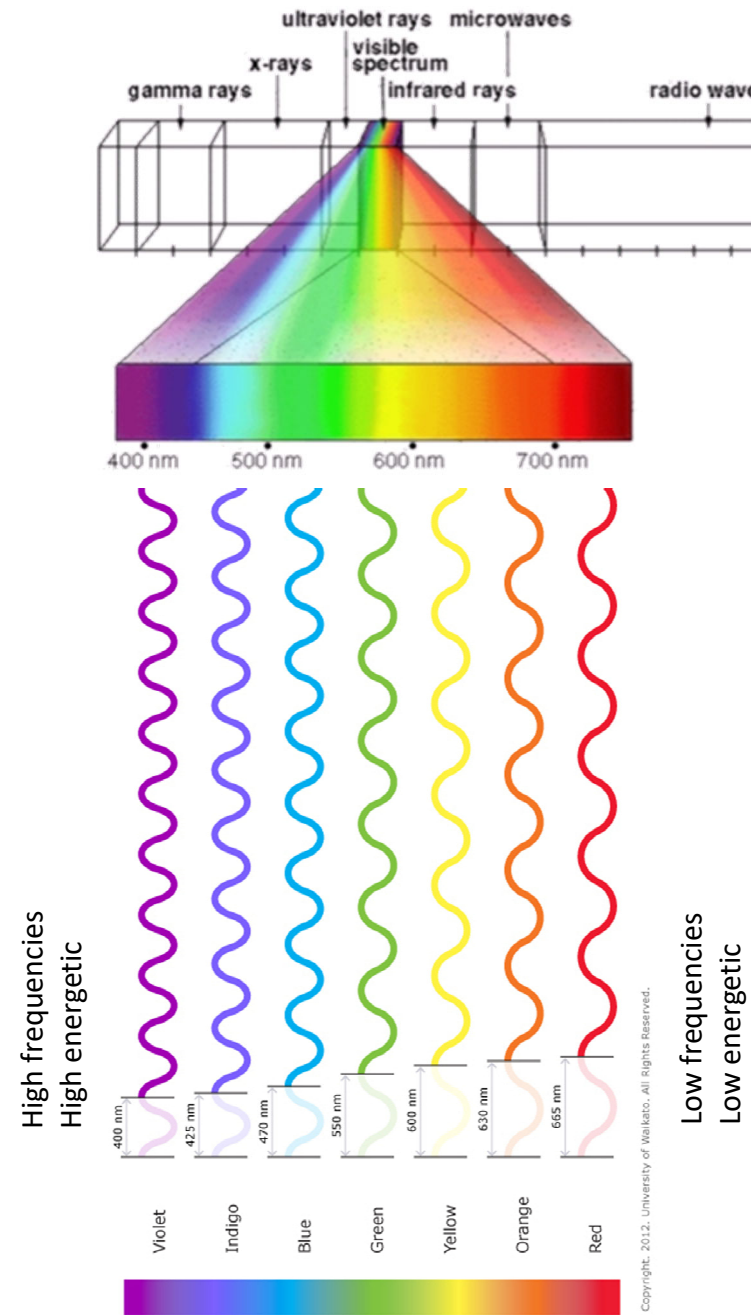
☐ Visual

- Image
 - Interpretation: colour

Note:

Colour is **NO** property of the physics of light

Colour is the human interpretation of different energetic frequencies (short waves, long waves) that hit a cell



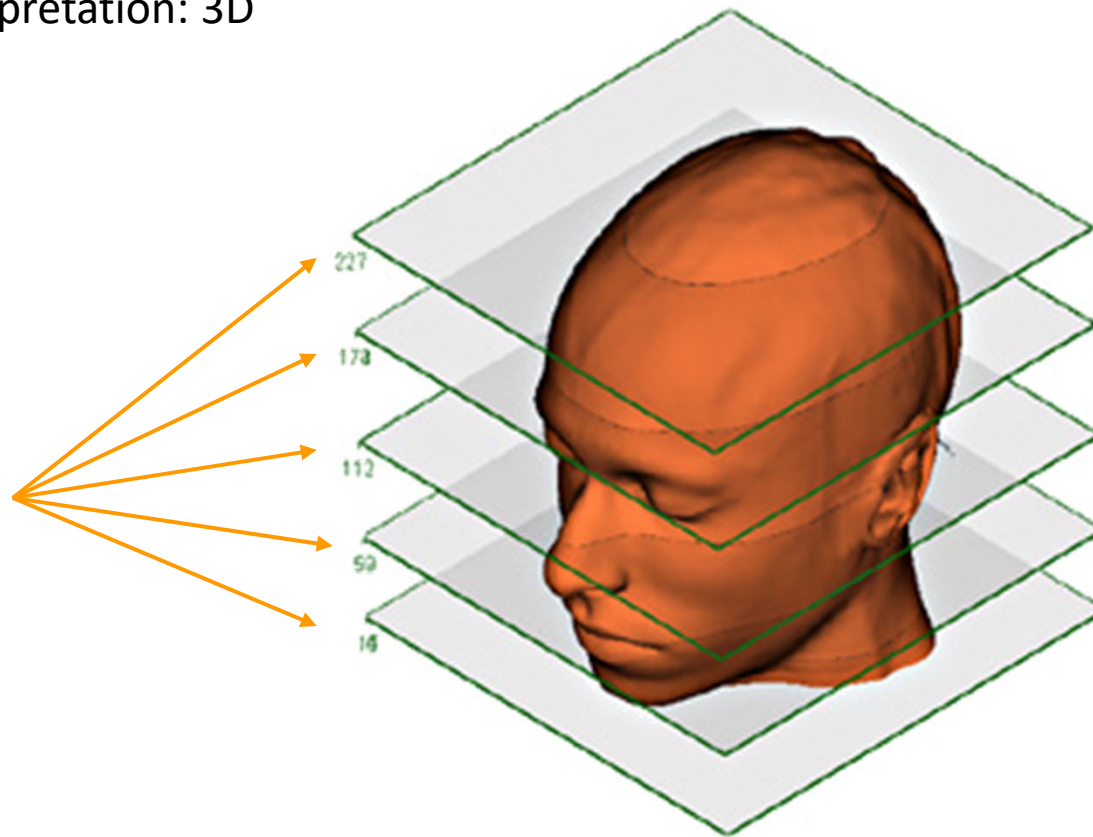
Input: perception

☐ Visual

- Image
 - Interpretation: 3D

CT scanner slices up an object
into several layers

Computed
Tomography
Imaging
(CT Scan,
CAT Scan)



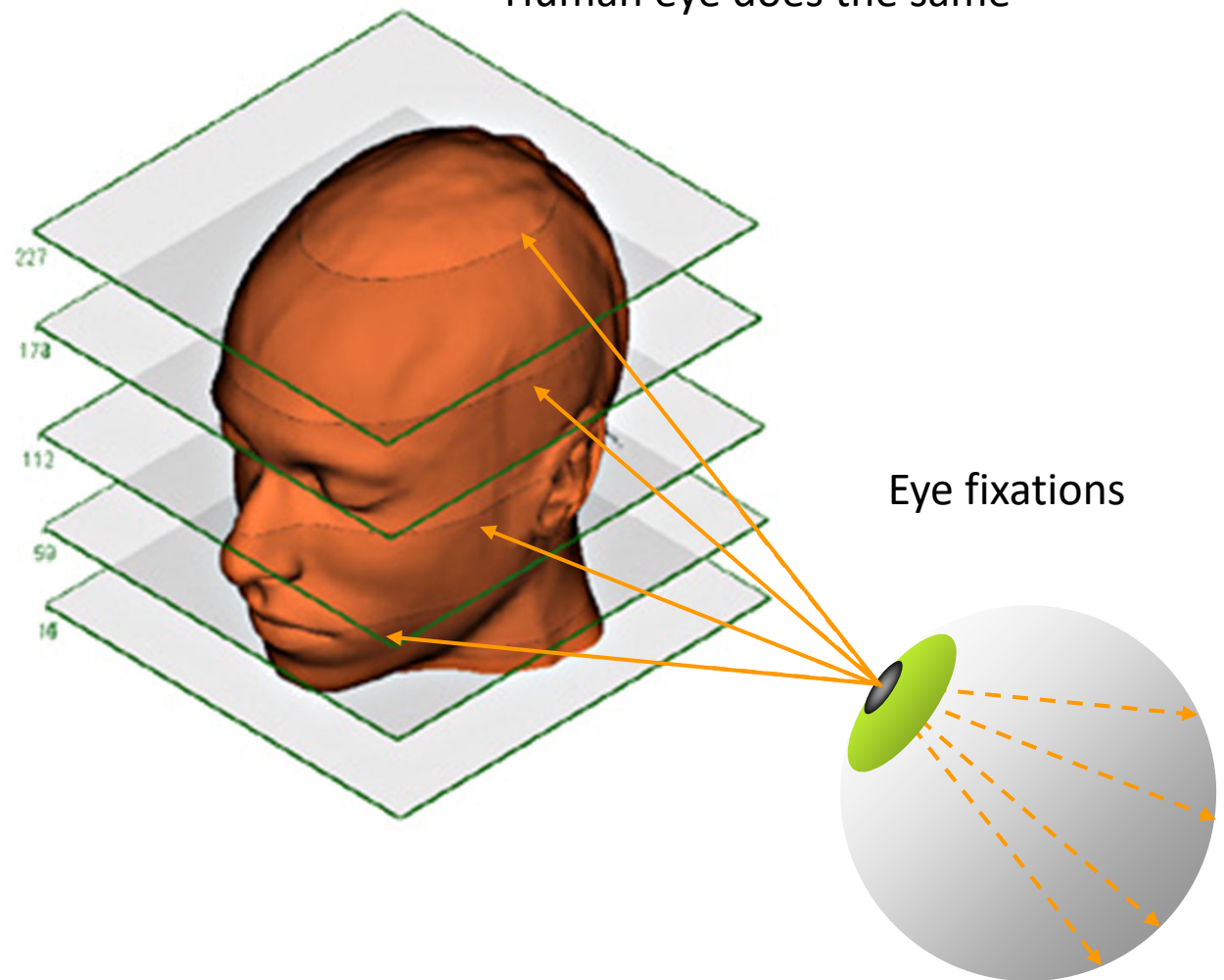
Input: perception

□ Visual

- Image
- Interpretation: 3D

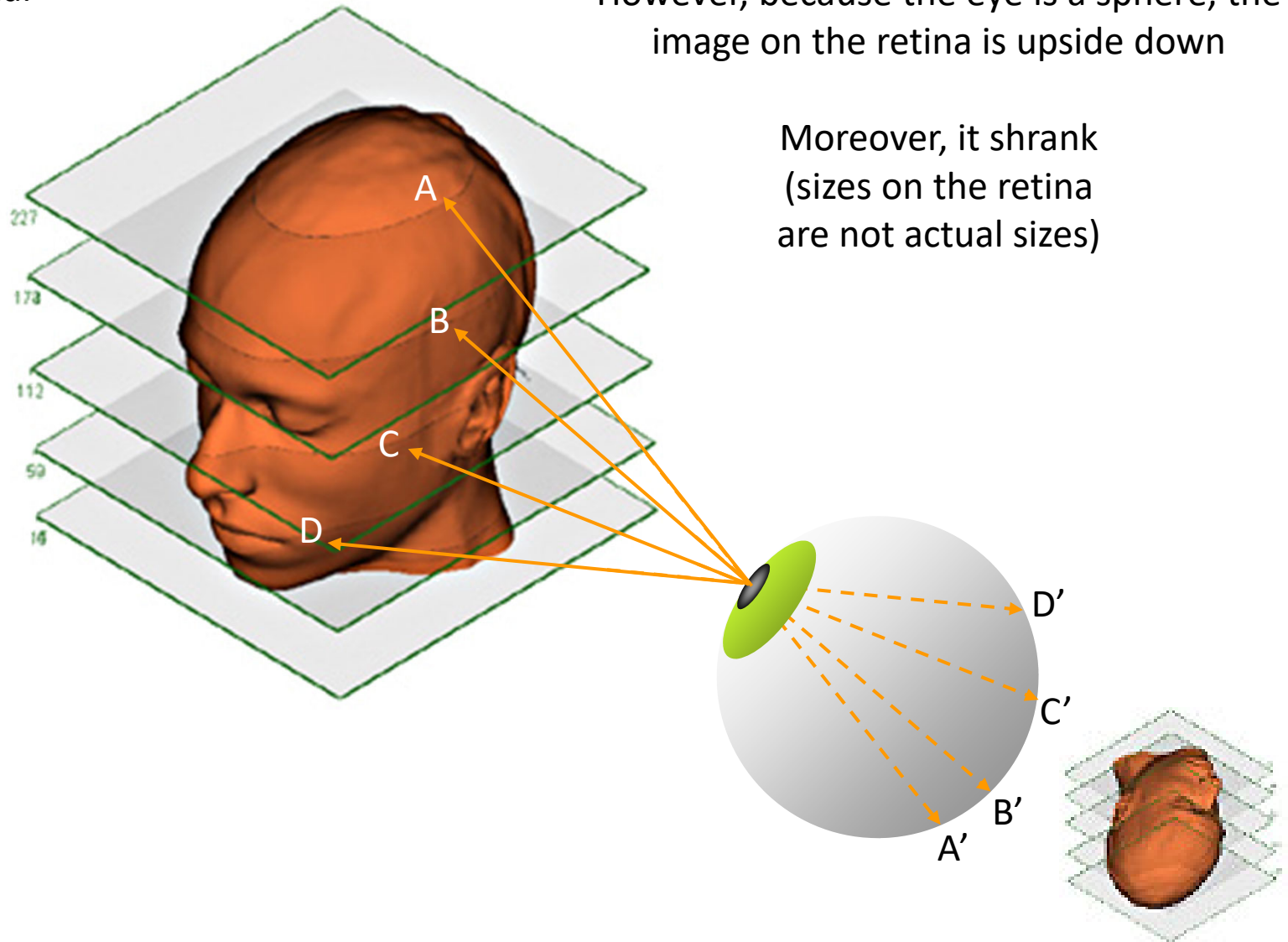
Missing data (in
between the
fixations) are filled
in conceptually
(from memory)

CT scans are slices of an object,
showing several layers
Human eye does the same



Input: perception

☐ Visual



Input: perception

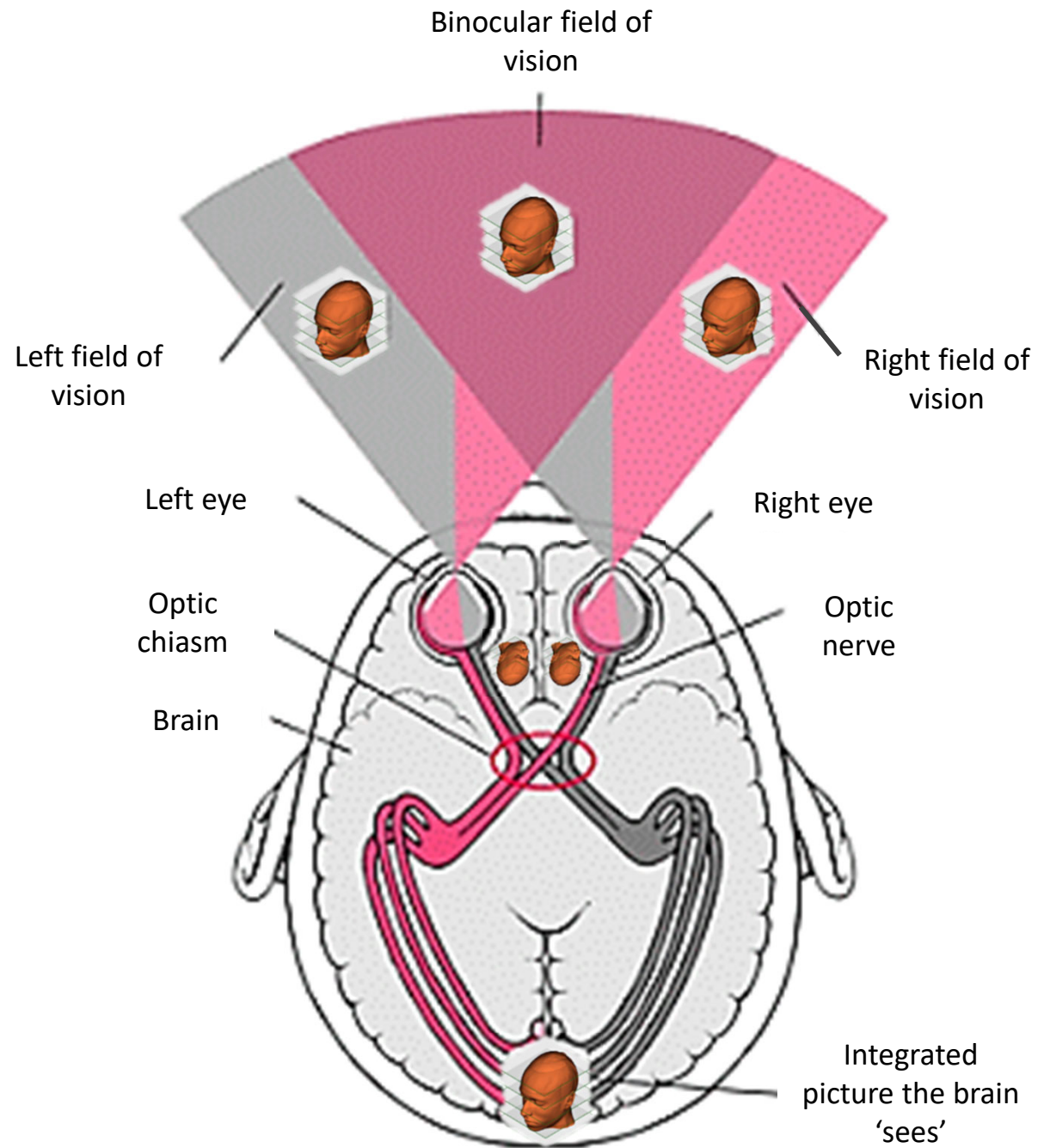
☐ Visual

Mind you, you have two eyes
So there are two upside down images

These two images were taken from different angles

To merge them into one, they enter the optic chiasm

Images are split:
left goes right
right goes left



Input: perception

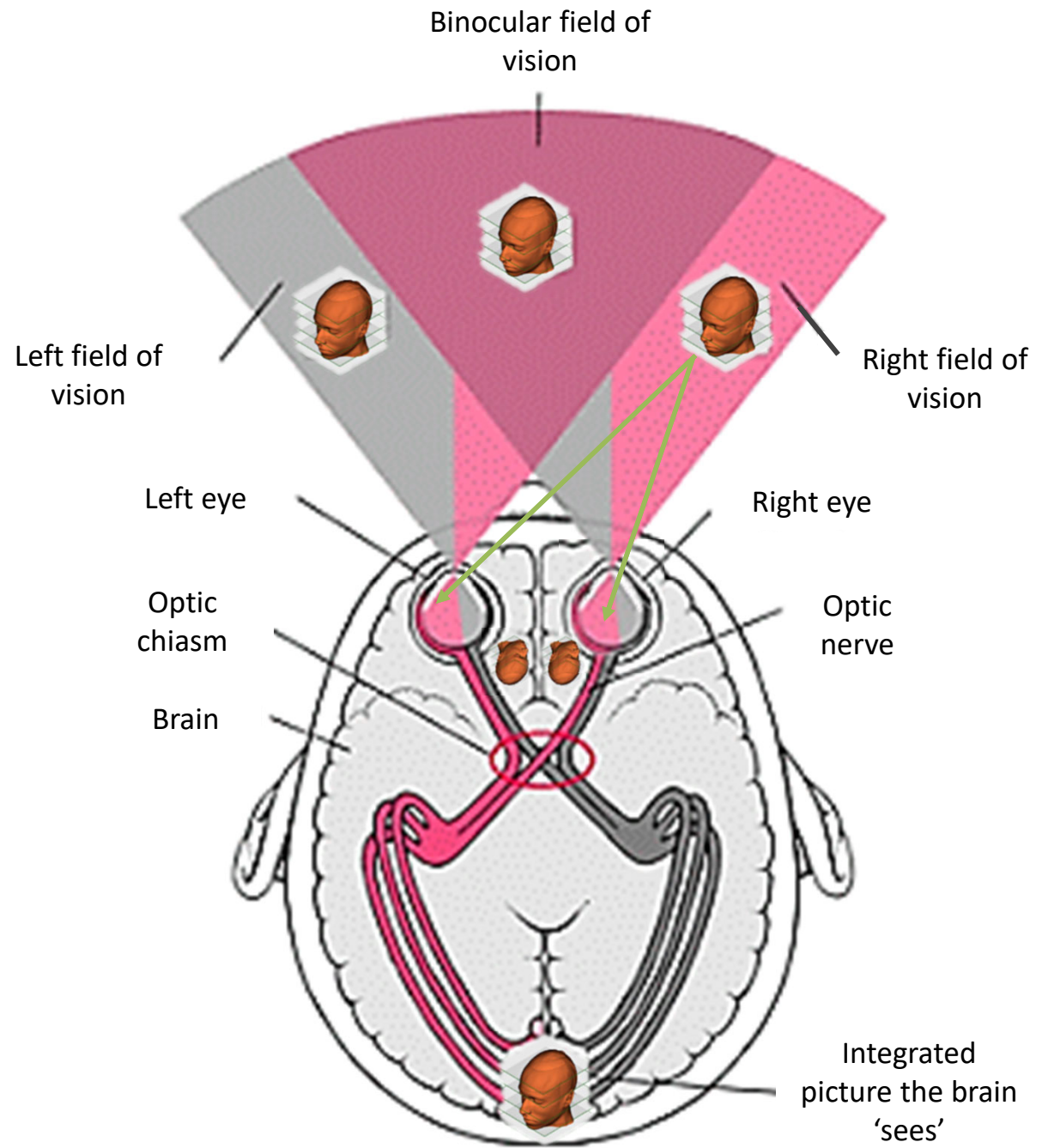
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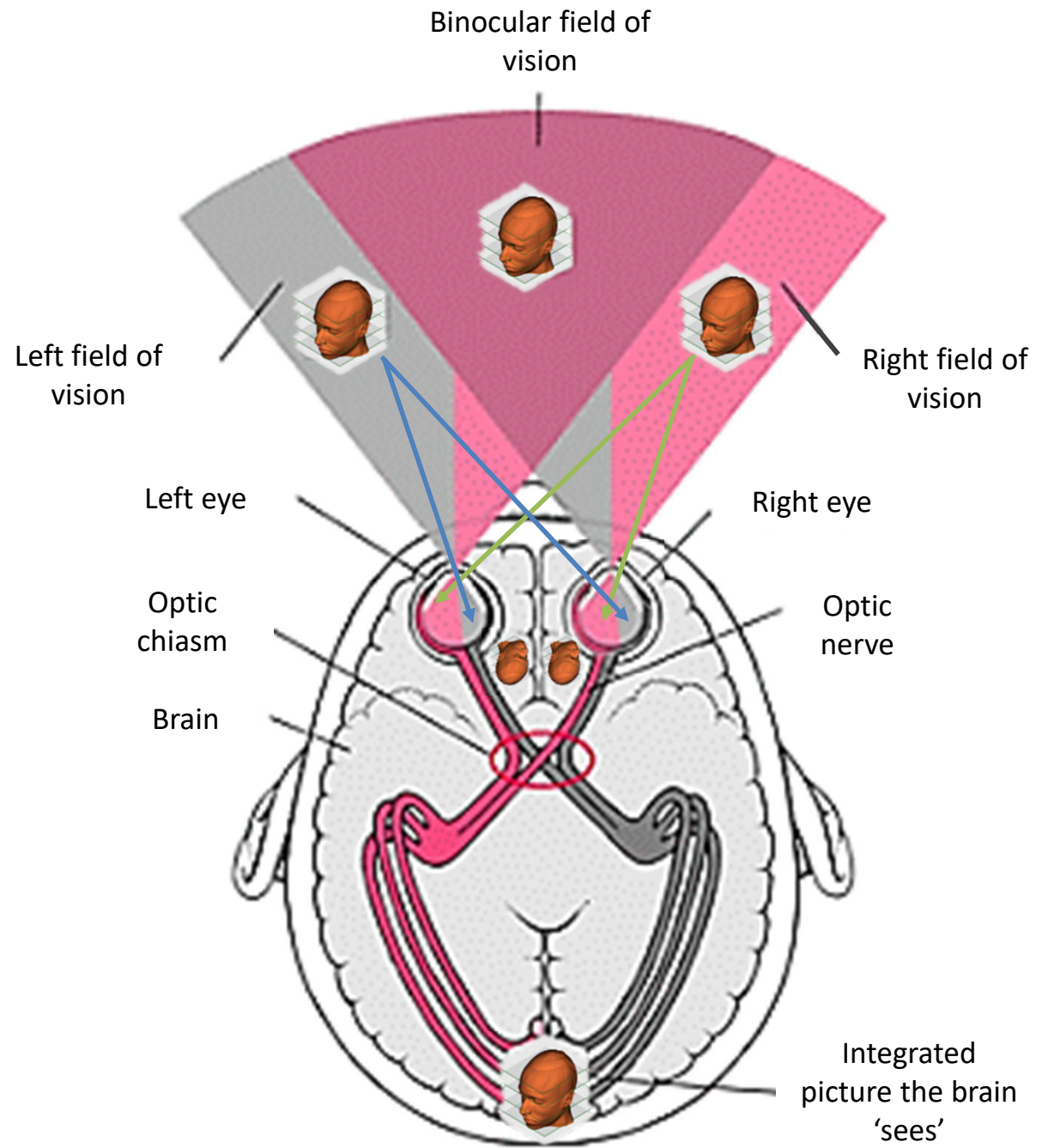
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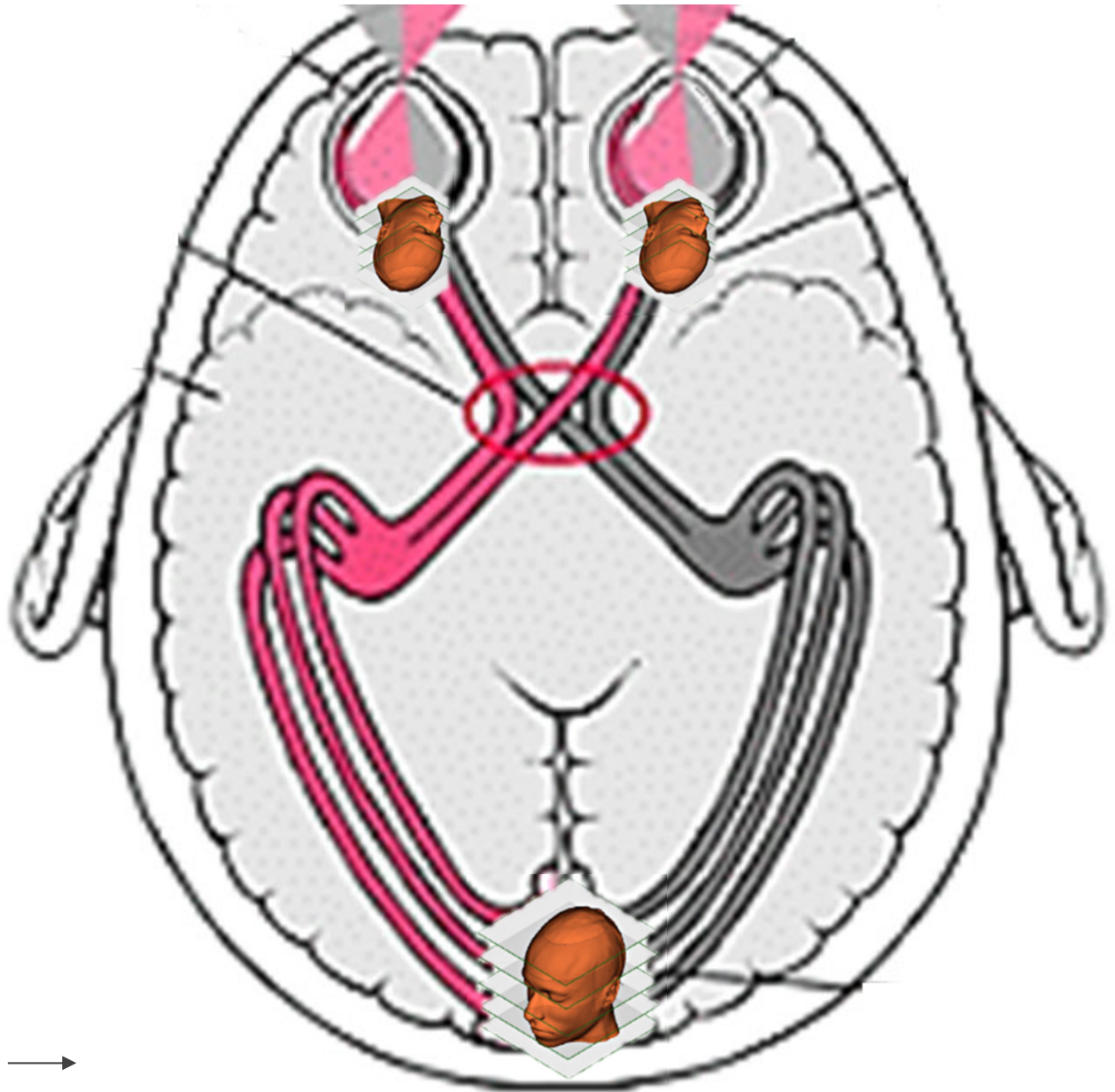
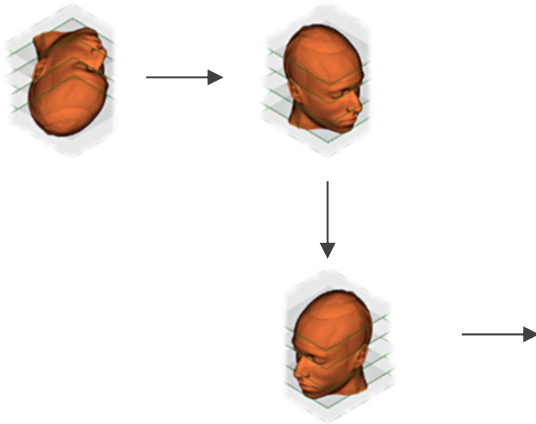


Input: perception

☐ Visual

But before the image can be passed on from the visual cortex to the next station, it must be:

Flipped, rotated,
and resized

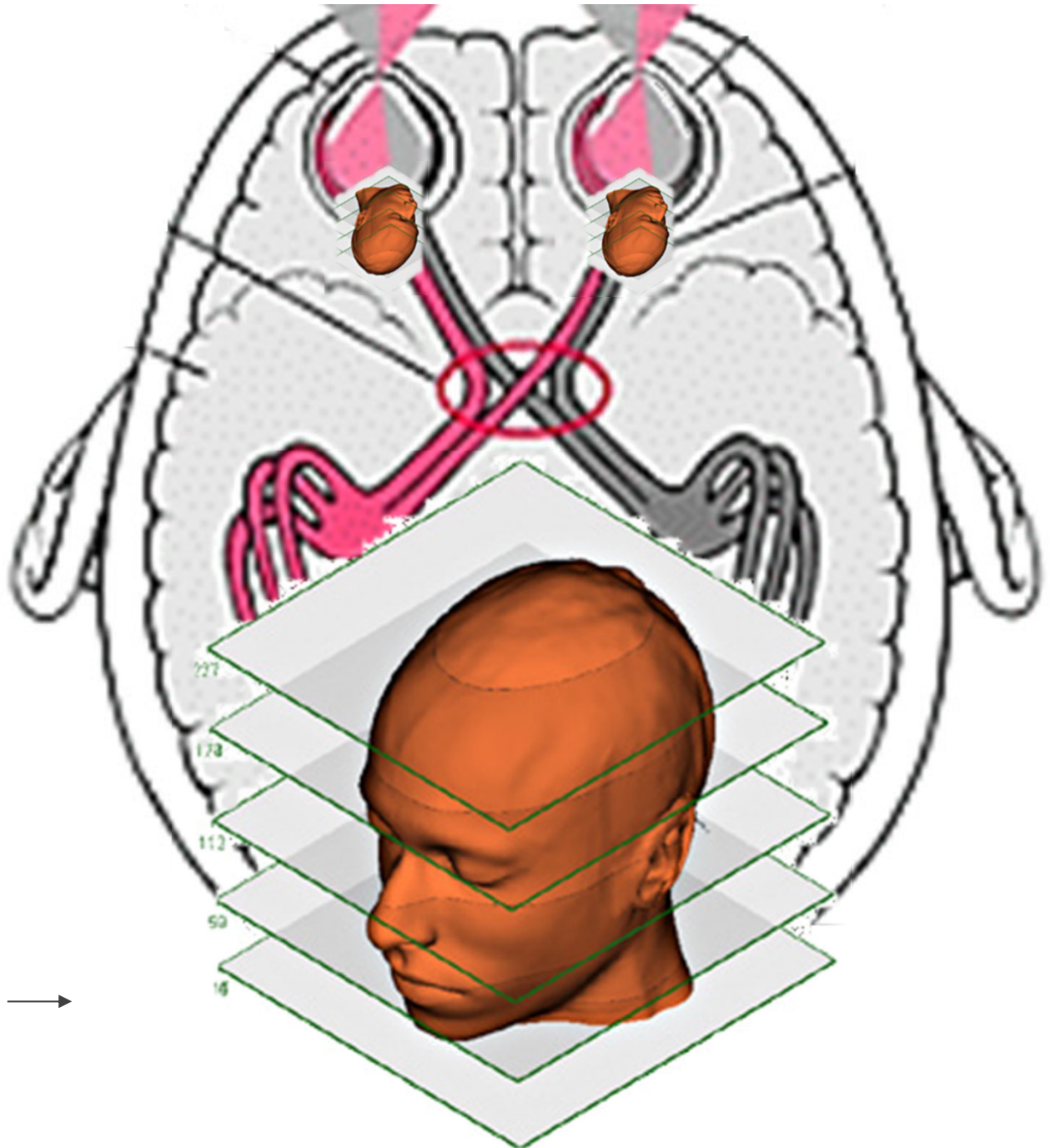
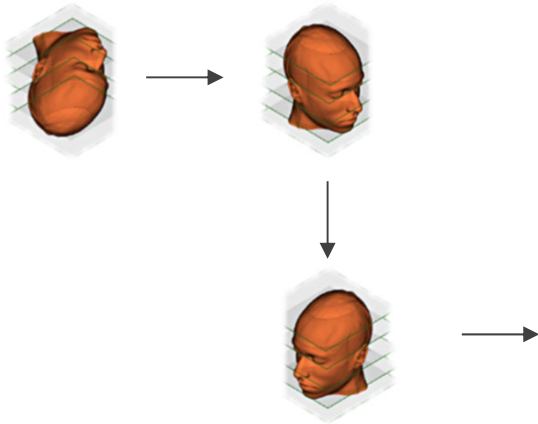


Input: perception

☐ Visual

But before the image can be passed on from the visual cortex to the next station, it must be:

Flipped, rotated,
and resized



Input: perception

☐ Visual

- Image
 - Interpretation: 3D



Therefore, 3D vision is **NO** property of image on the retina (which is 2D)

- Slicing up (eye fixations)
- Imputing missing data from memory (concept-driven percept)
- Shrinking images
- Taking them from different angles
- Splitting them up
- Left right, right left
- Merging
- Flip
- Rotate
- Resize (blow up)



Delvcam DELV-3DMS Dual 8 Inch LCD Monitor with Attached 3D Stereoscopic Viewer

Input: perception

❑ Visual

- Image
 - Interpretation: 3D

The illusion of depth is so strong that it urges our brain to resize objects so as to fit our conceptual interpretation of the environment

Size and depth



Giant or gnome?

Slanted lines indicate size through depth suggestion

Input: perception

☐ Visual

- Image
 - Interpretation: 3D

Size and depth

- Visual angle indicates how much of view is occupied by the object
- Related to size and distance from eye
- Visual acuity is ability to perceive fine detail
- 'Law of size constancy' means that familiar objects are perceived as constant size in spite of changes in visual angle
- Cues like 'overlapping' help perception of size and depth

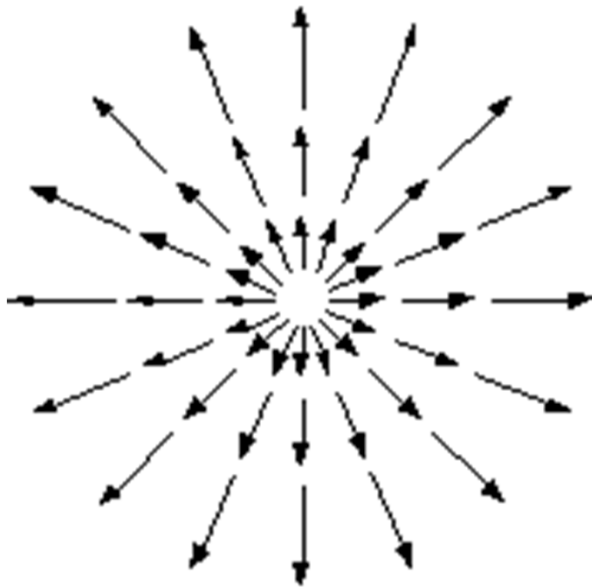


Input: perception

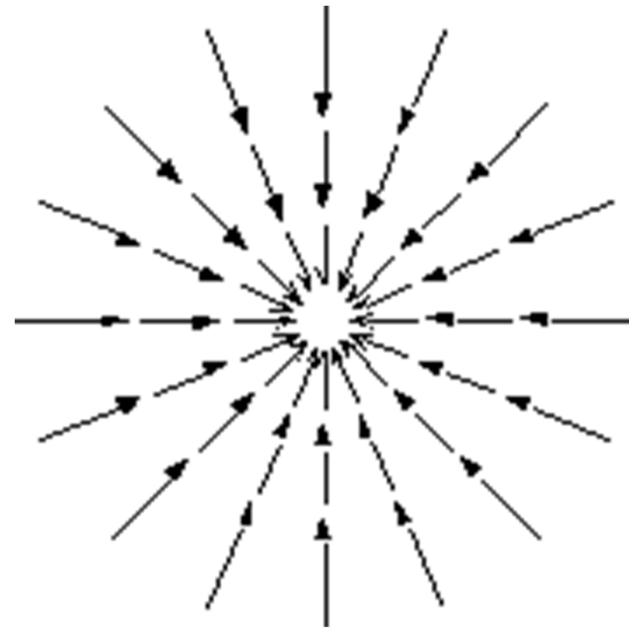
☐ Visual

- Image
 - Interpretation: 3D

Positioning through motion



Expanding optic flow makes the body
lean forward



Contracting optic flow makes the body
lean backward

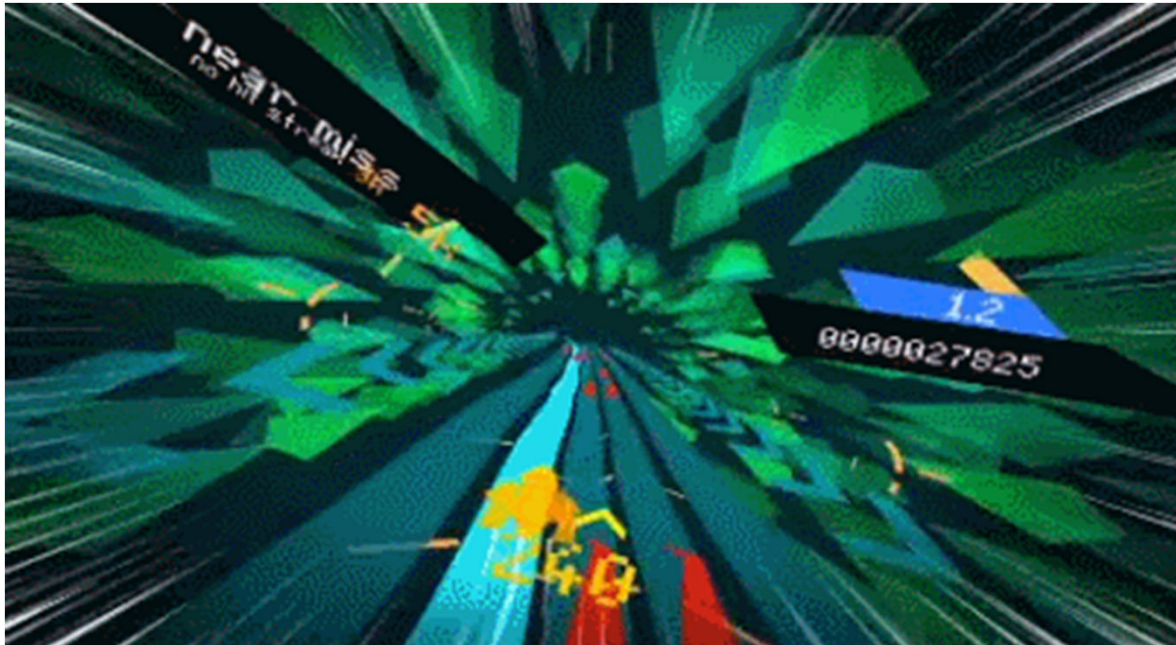
Input: perception

☐ Visual

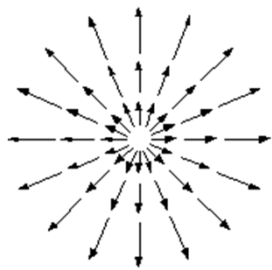
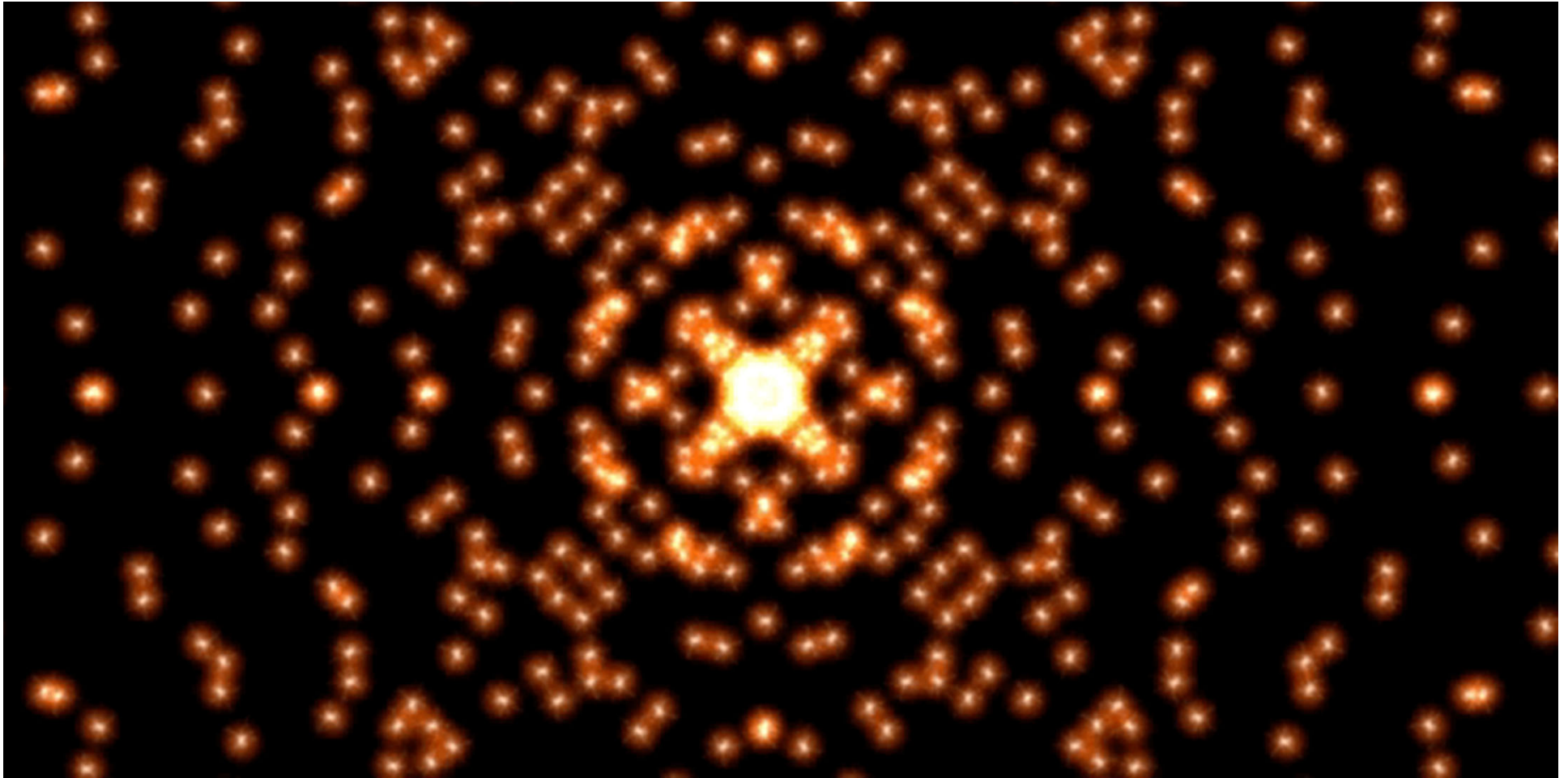
- Image
 - Interpretation: 3D

Optic flow is the brain's cue to calculate position in space

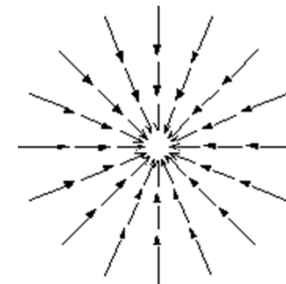
Positioning through motion



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lean forward



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lean forward



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lean backward

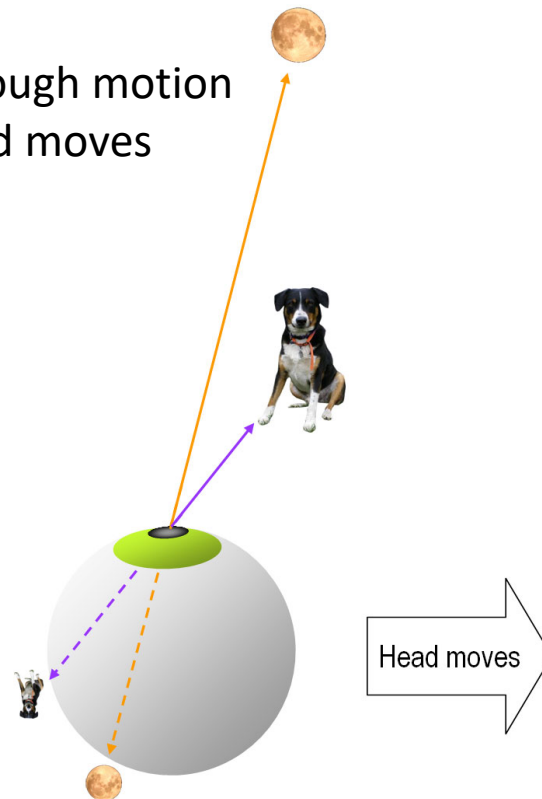
Input: perception

❑ Visual

- Image

- Interpretation: 3D

Seeing 3D through motion
when the head moves



Dog 'moves' more on the retina than the moon because the length of the projection path of the dog on the retina is longer (purple arrow) than that of the moon (orange arrow)

The brain interprets this as: Short path is behind long path; therefore, there must be depth

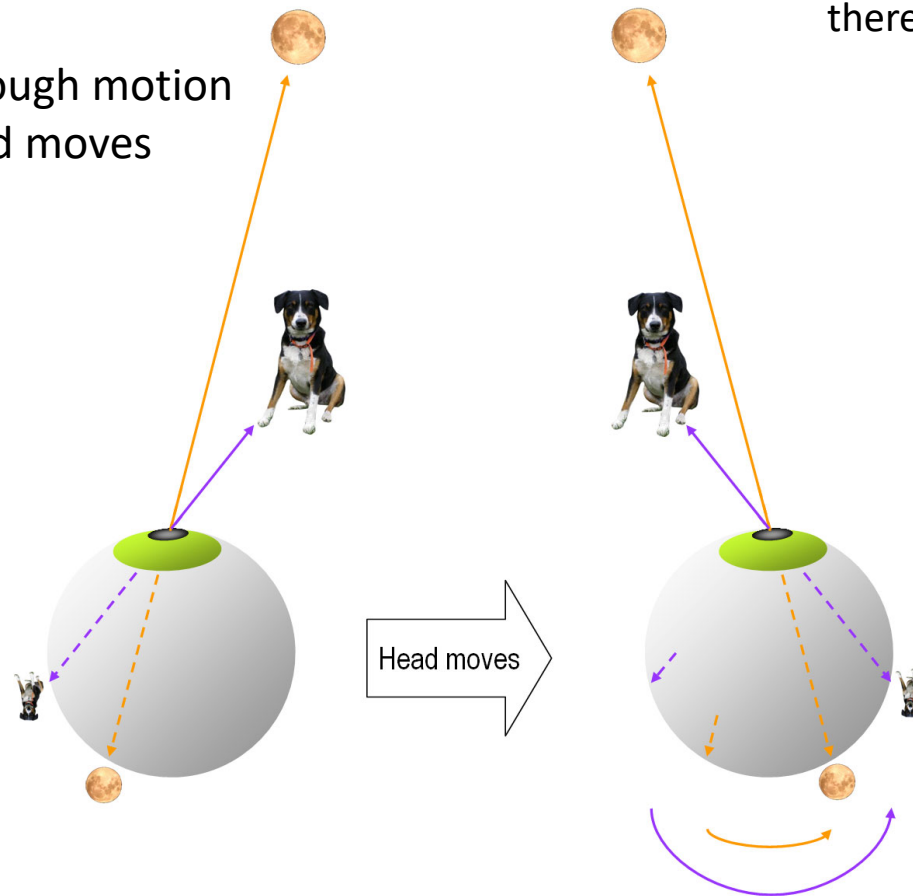
Input: perception

□ Visual

• Image

- Interpretation: 3D

Seeing 3D through motion
when the head moves



Dog 'moves' more on the retina than the moon
because the length of the projection path of the dog
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the moon (orange arrow)

The brain interprets this as: short
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therefore, there must be depth

Input: perception

□ Visual

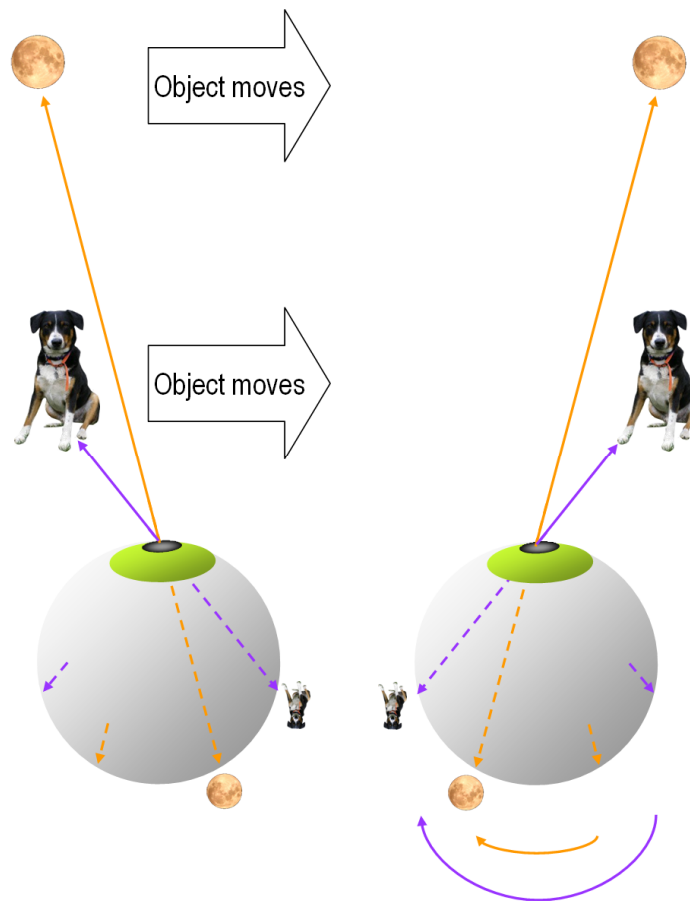
- Image

- Interpretation: 3D

The object closest to the eye 'moves' the most because the retinal path is longer. The brain interprets this as: short path is behind long path; therefore, there must be depth

Seeing 3D through motion when the object moves

Therefore,
'Moving-train illusion'
It's the other train that's moving

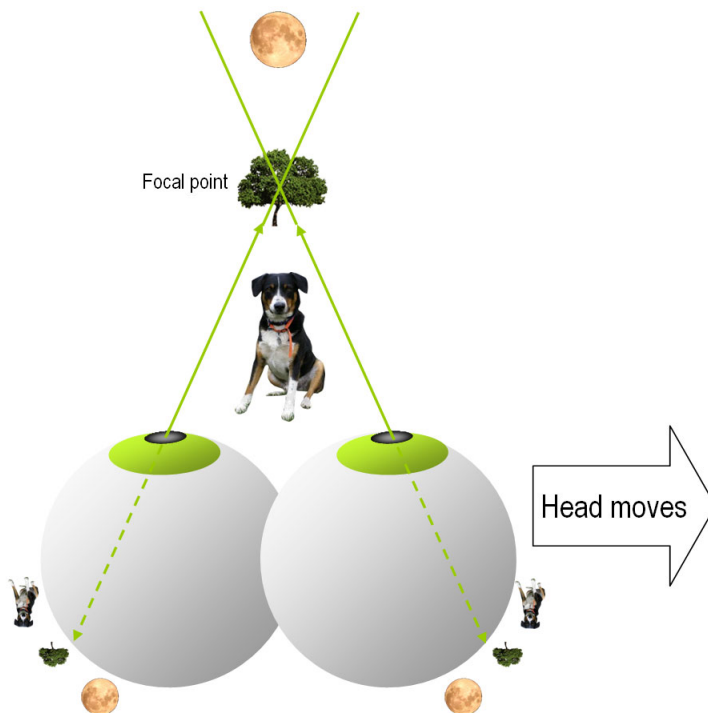


Input: perception

☐ Visual

- Image
 - Interpretation: 3D

Motion parallax



The direction in which an object leaves the visual field indicates the direction of object motion in the physical world. Moon travels with you (which may not be true)

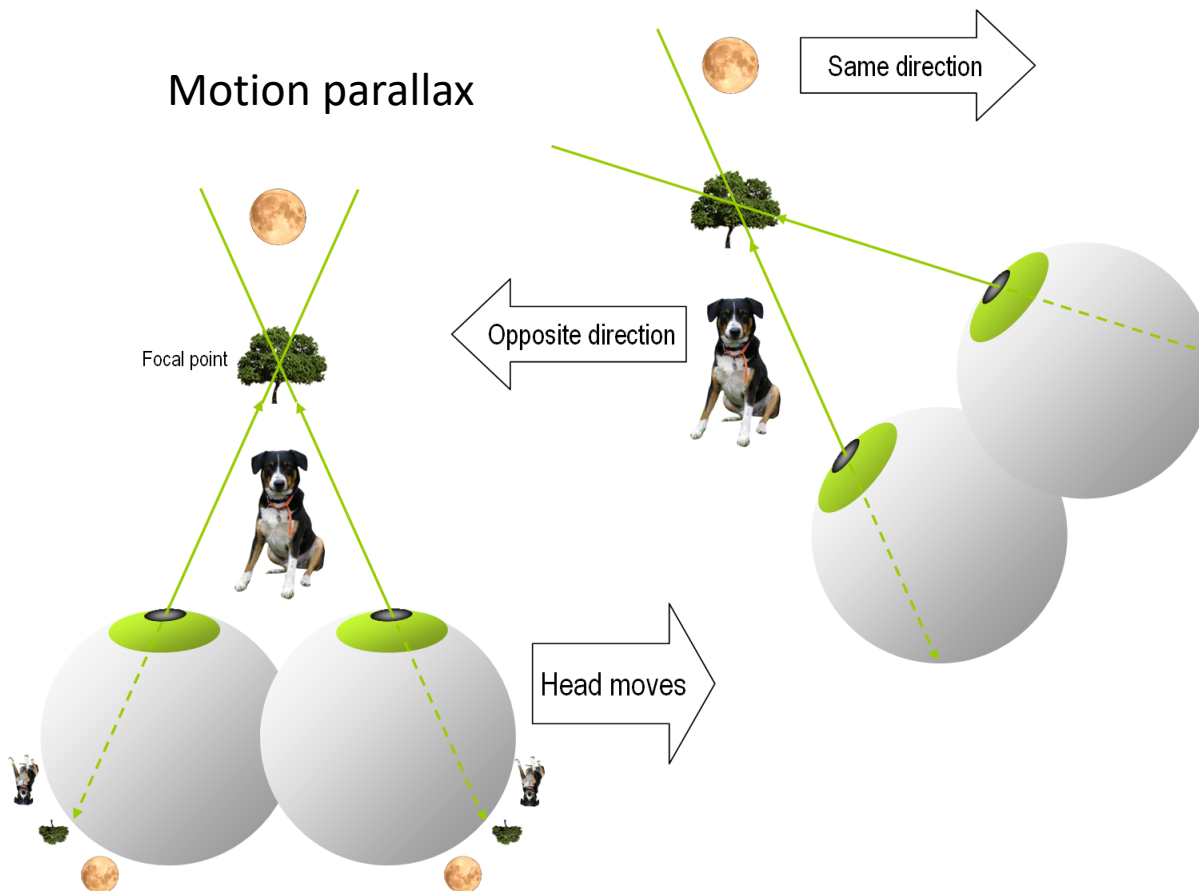
Thus, direction of leaving the visual half-field cues object motion and despite conceptually knowing that the moon does not fly with you, the simple everyday heuristics make us see it that way

Input: perception

□ Visual

- Image
- Interpretation: 3D

Motion parallax



The direction in which an object leaves the visual field indicates the direction of object motion in the physical world. Moon travels with you (which may not be true)

Thus, direction of leaving the visual half-field cues object motion and despite conceptually knowing that the moon does not fly with you, the simple everyday heuristics make us see it that way

Input: perception

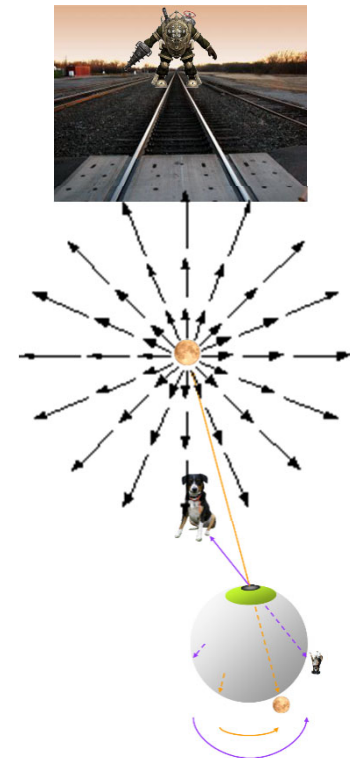
❑ Visual

- Image
 - Interpretation: 3D



Therefore, 3D vision is **NO** property of image on the retina (which is 2D)

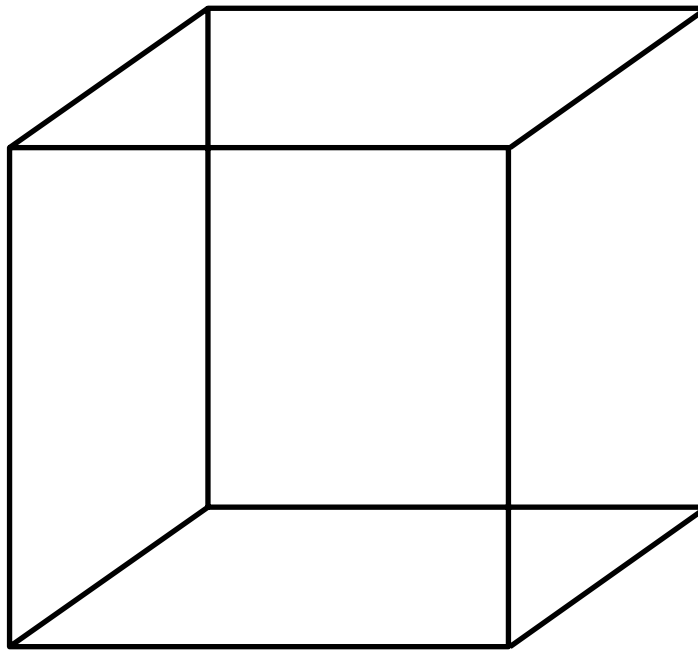
- Visual angle indicates object size
- Related to size and distance from eye
- Size constancy despite changes in visual angle
- Expanding optic flow indicates 'we go forward'
- Contracting optic flow indicates 'backward'
- Long retinal path indicates 'close by'
- Short retinal path indicates 'far away'
- Side of leaving visual half field indicates motion direction:
 - side of your direction means object goes with me
 - opposite side of your direction means object stays behind



Input: perception

☐ Visual

- What do you see?



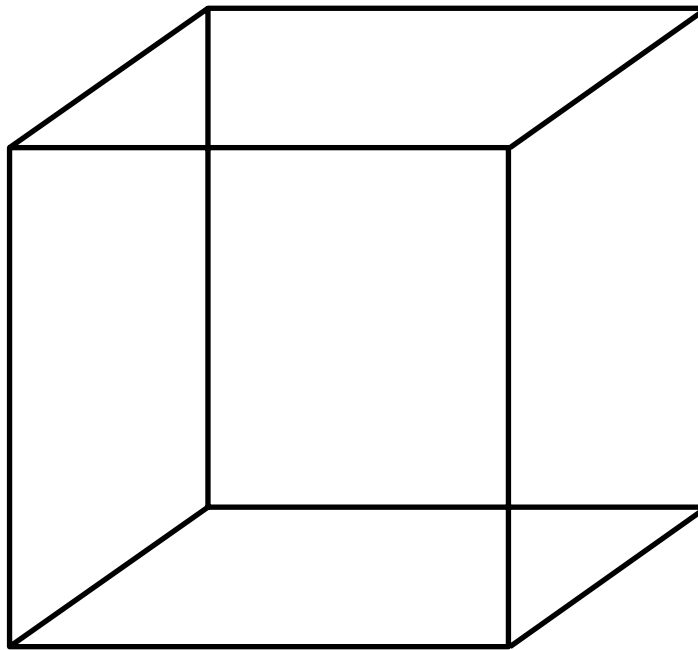
It's not a cube

Input: perception

☐ Visual

- The designer never drew a cube

1 cube =
2 right triangles
1 rectangle
4 trapezoids

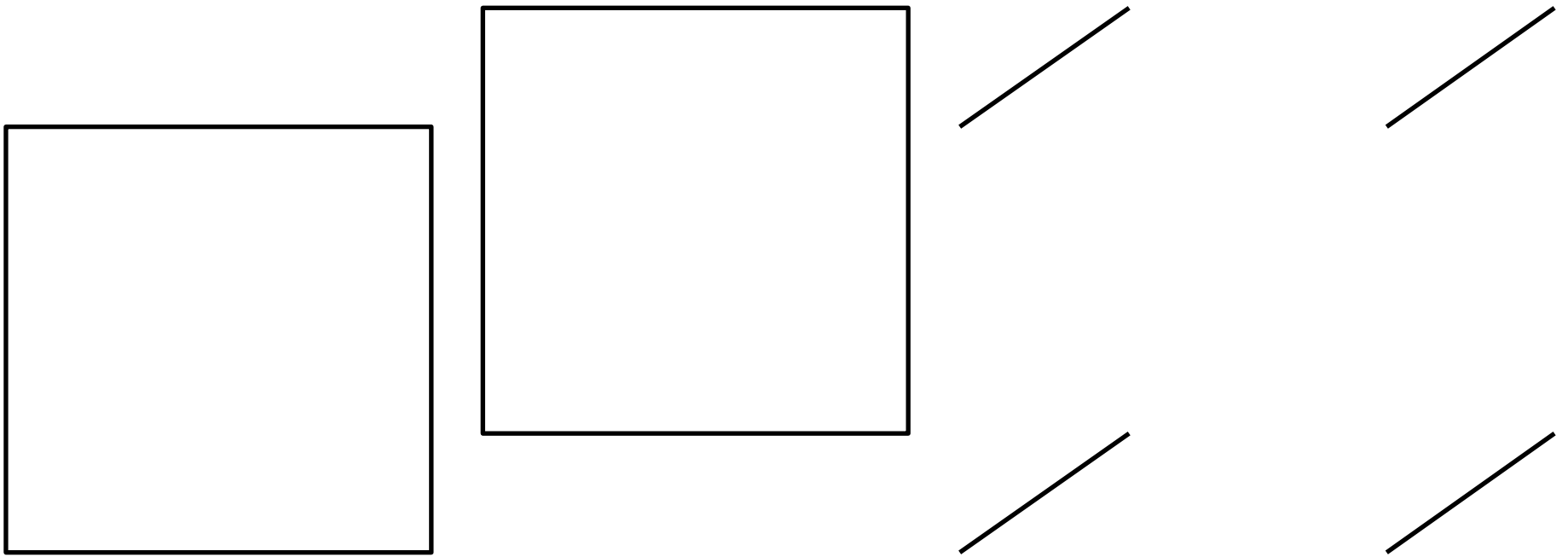


And even that is not true

Input: perception

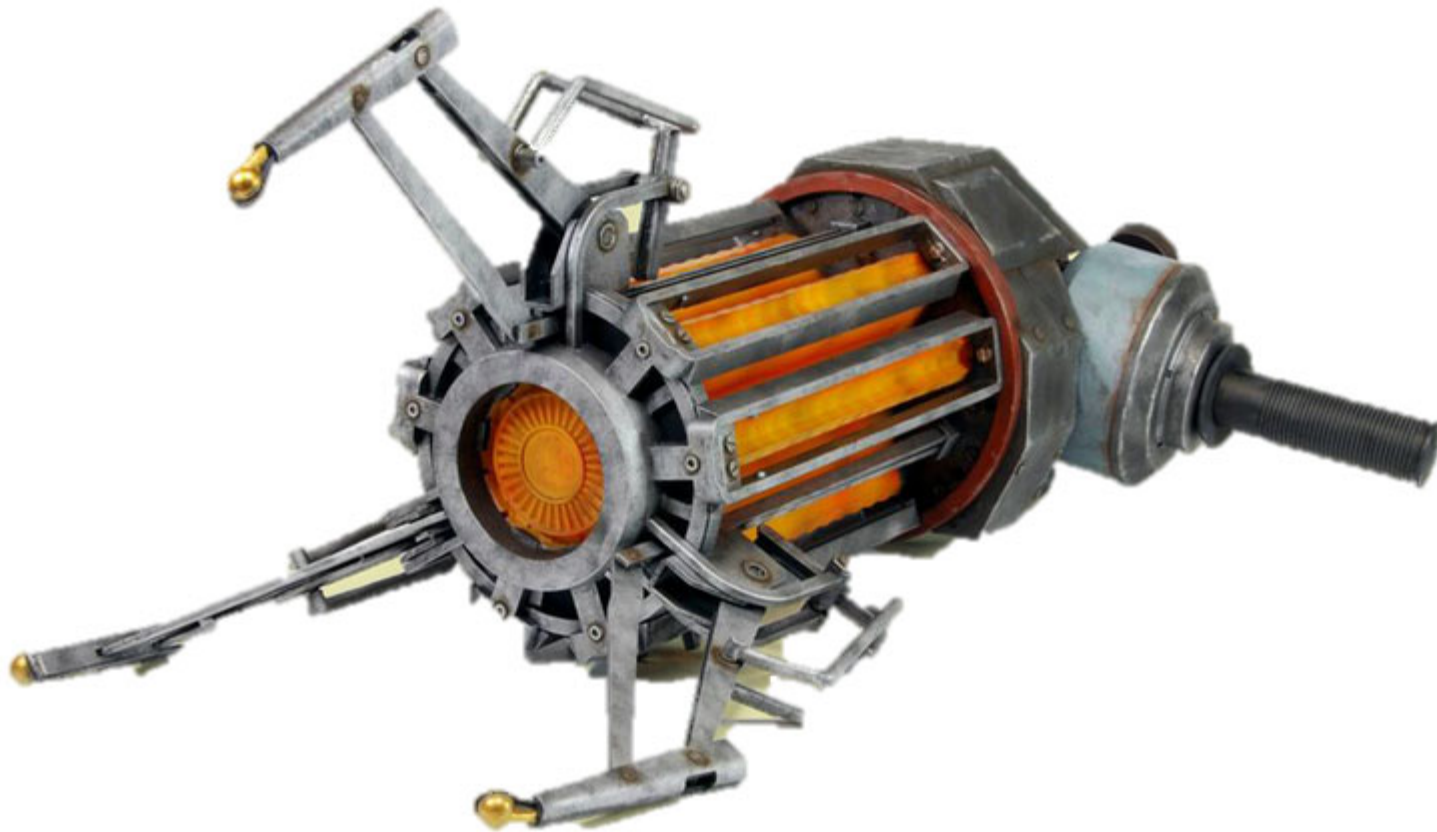
☐ Visual

- This is what the designer drew:



2 squares and 4 lines and the cube was not transparent

So what is 3D about this?



Or this?



Overwatch

Or this?



Next Car Game

Short break, after that:
Pattern recognition

Pattern recognition

Gestalt principles of perception

Input: perception

☐ Visual

- Image
 - Pattern recognition

	A	B	C	D	E	F	G	H	I	J
1		Apple1	MS1	Apple2	MS2	Apple3	MS3	Apple4	MS4	
2	Chin Ho	8	1	5	1	9	1	7	1	
3	Po Ki	2	3	1	2	2	3	1	2	
4	King to	7	1	9	1	7	1	8	2	
5	Jiayun	1	2	2	3	1	3	2	3	
6	Yu Heng	5	1	7	1	9	2	9	1	
7	Jin	1	2	1	3	2	3	2	3	
8	Genggeng	9	2	9	2	8	1	6	1	
9	Mingyi	2	3	1	2	1	2	2	3	
10	Yuyang	8	1	9	1	8	2	4	1	
11	Yajie	1	2	1	3	1	3	2	3	
12	Yifan	8	2	7	1	7	1	9	1	
13	Xianghui	1	3	2	3	1	2	1	3	
14										
15										

Which is higher, which is lower?

Input: perception

☐ Visual

- Image
 - Pattern recognition

	A	B	C	D	E	F	G	H	I	J
1		Apple1	MS1	Apple2	MS2	Apple3	MS3	Apple4	MS4	
2	Chin Ho	8	1	5	1	9	1	7	1	
3	Po Ki	2	3	1	2	2	3	1	2	
4	King to	7	1	9	1	7	1	8	2	
5	Jiayun	1	2	2	3	1	3	2	3	
6	Yu Heng	5	1	7	1	9	2	9	1	
7	Jin	1	2	1	3	2	3	2	3	
8	Genggeng	9	2	9	2	8	1	6	1	
9	Mingyi	2	3	1	2	1	2	2	3	
10	Yuyang	8	1	9	1	8	2	4	1	
11	Yajie	1	2	1	3	1	3	2	3	
12	Yifan	8	2	7	1	7	1	9	1	
13	Xianghui	1	3	2	3	1	2	1	3	
14		high	low	high	low	high	low	high	low	
15										

Which is higher, which is lower?

Input: perception

❑ Visual

- Image
 - Pattern recognition

	A	B	C	D	E	F	G	H	I	J
1		Apple1	MS1	Apple2	MS2	Apple3	MS3	Apple4	MS4	
2	Chin Ho	high	low	high	low	high	low	high	low	
3	Po Ki	low	high	low	high	low	high	low	high	
4	King to	high	low	high	low	high	low	high	low	
5	Jiayun	low	high	low	high	low	high	low	high	
6	Yu Heng	high	low	high	low	high	low	high	low	
7	Jin	low	high	low	high	low	high	low	high	
8	Genggeng	high	low	high	low	high	low	high	low	
9	Mingyi	low	high	low	high	low	high	low	high	
10	Yuyang	high	low	high	low	high	low	high	low	
11	Yajie	low	high	low	high	low	high	low	high	
12	Yifan	high	low	high	low	high	low	high	low	
13	Xianghui	low	high	low	high	low	high	low	high	
14										

Which is higher, which is lower?

Input: perception

❑ Visual

- Image
 - Pattern recognition

	A	B	C	D	E	F	G	H	I	J
1		Apple1	MS1	Apple2	MS2	Apple3	MS3	Apple4	MS4	
2	Male	high	low	high	low	high	low	high	low	
3	Female	low	high	low	high	low	high	low	high	
4	Male	high	low	high	low	high	low	high	low	
5	Female	low	high	low	high	low	high	low	high	
6	Male	high	low	high	low	high	low	high	low	
7	Female	low	high	low	high	low	high	low	high	
8	Male	high	low	high	low	high	low	high	low	
9	Female	low	high	low	high	low	high	low	high	
10	Male	high	low	high	low	high	low	high	low	
11	Female	low	high	low	high	low	high	low	high	
12	Male	high	low	high	low	high	low	high	low	
13	Female	low	high	low	high	low	high	low	high	
14										

Which is higher, which is lower?

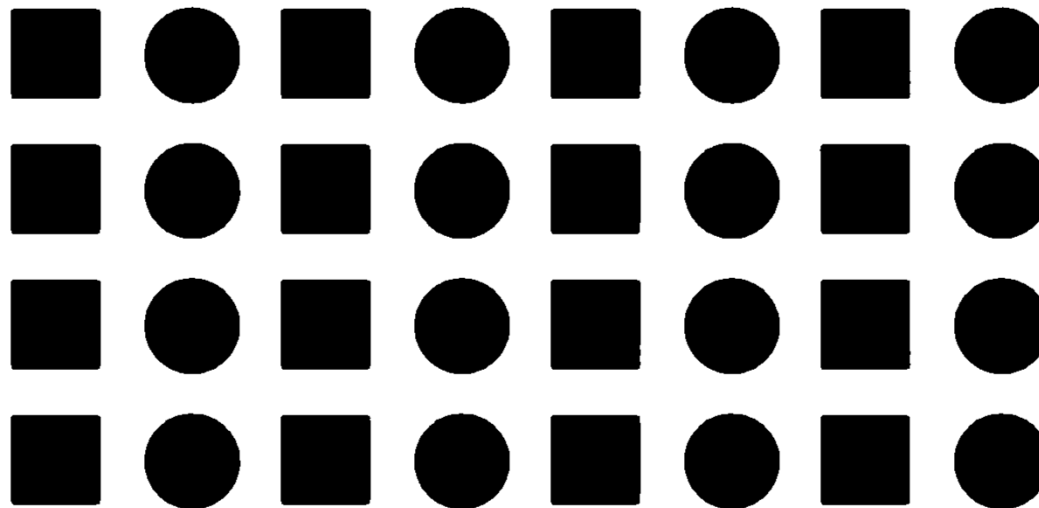
Input: perception

□ Visual

Items that are similar tend to be grouped together

- Image

- Pattern recognition: similarity



Most people will see columns of squares and circles, rather than rows of alternating shapes

Input: perception

☐ Visual

- Image
 - Pattern recognition

	A	B	C	D	E	F	G
1		var1	var2	var3	var4	var5	
2	record1	3	3	7	7	7	
3	record2	3	3	7	7	7	
4	record3	3	3	7	7	7	
5	record4	3	3	0	0	0	
6	record5	3	3	0	0	0	
7	record6	3	3	0	0	0	
8	record7	3	3	0	0	0	
9							
10							

Do you see any grouping in this data set?

Input: perception

❑ Visual

- Image
 - Pattern recognition

Humans tend to recognise
patterns even where they are
not

	A	B	C	D	E	F	G
1		var1	var2	var3	var4	var5	
2	record1	3	3	7	7	7	
3	record2	3	3	7	7	7	
4	record3	3	3	7	7	7	
5	record4	3	3	0	0	0	
6	record5	3	3	0	0	0	
7	record6	3	3	0	0	0	
8	record7	3	3	0	0	0	
9							
10							

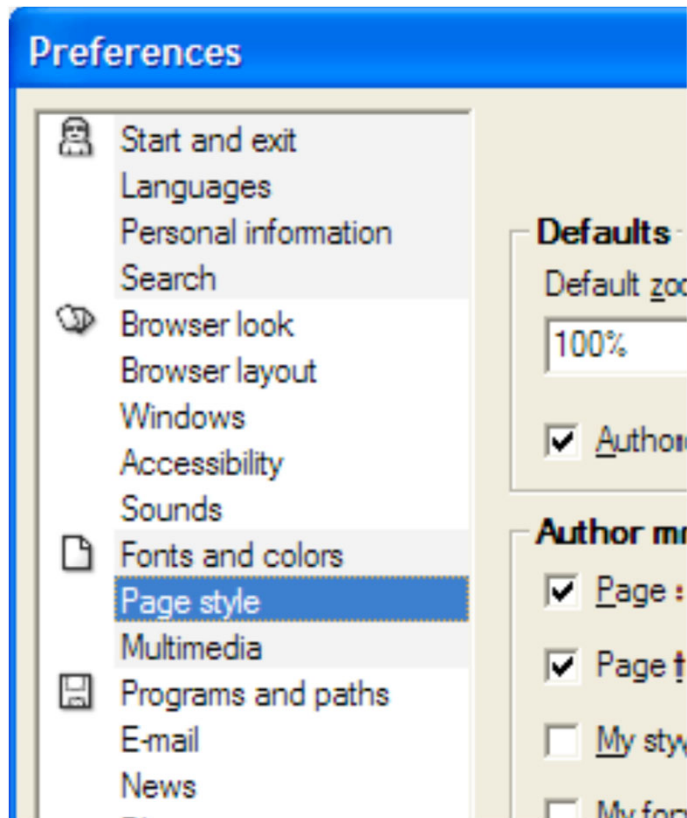
Probably you see three groups of data points
although these groups are unrelated

~~~~~

## Input: perception

### ☐ Visual

- Image
  - Pattern recognition: similarity



From the Opera browser.  
Options with the same  
background colour are  
“tied together”  
automatically by the user

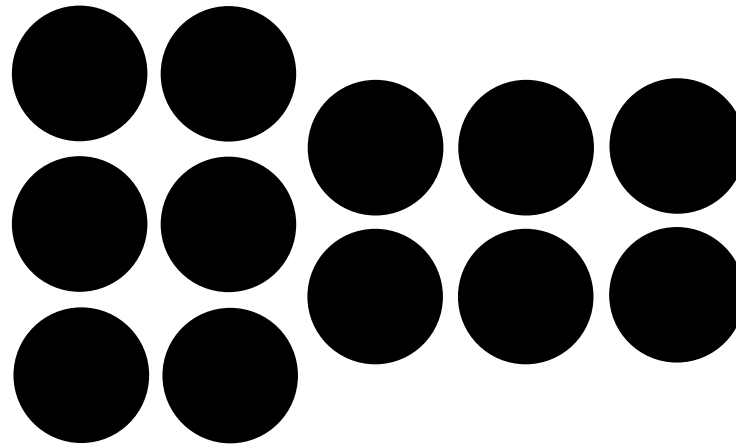
## Input: perception

□ Visual

Objects near each other tend to be grouped together

• Image

- Pattern recognition: proximity



We will most likely see the circles on the left as grouped into two columns, and those on the right as grouped into two rows

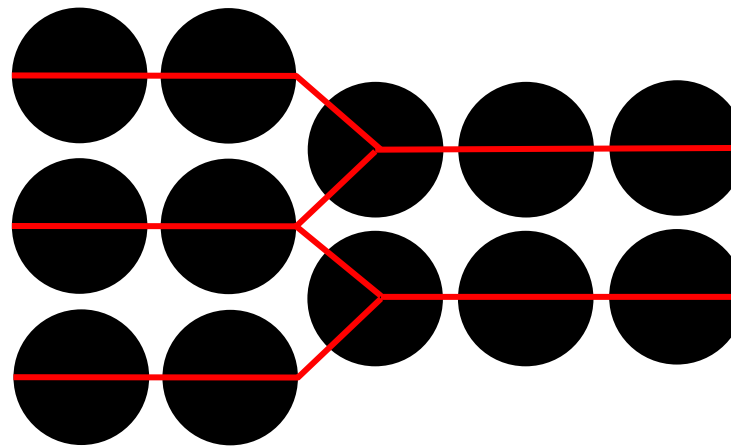
## Input: perception

☐ Visual

• Image

- Pattern recognition: proximity

Objects near each other tend to be grouped together



You probably did not see this one

We will most likely see the circles on the left as grouped into two columns, and those on the right as grouped into two rows

## Input: perception

☐ Visual

- Image

- Pattern recognition: proximity



☒ P2P Search ☐ Web Search

☒ Everything

☐ Audio

☐ Video

☐ Images

☐ Documents

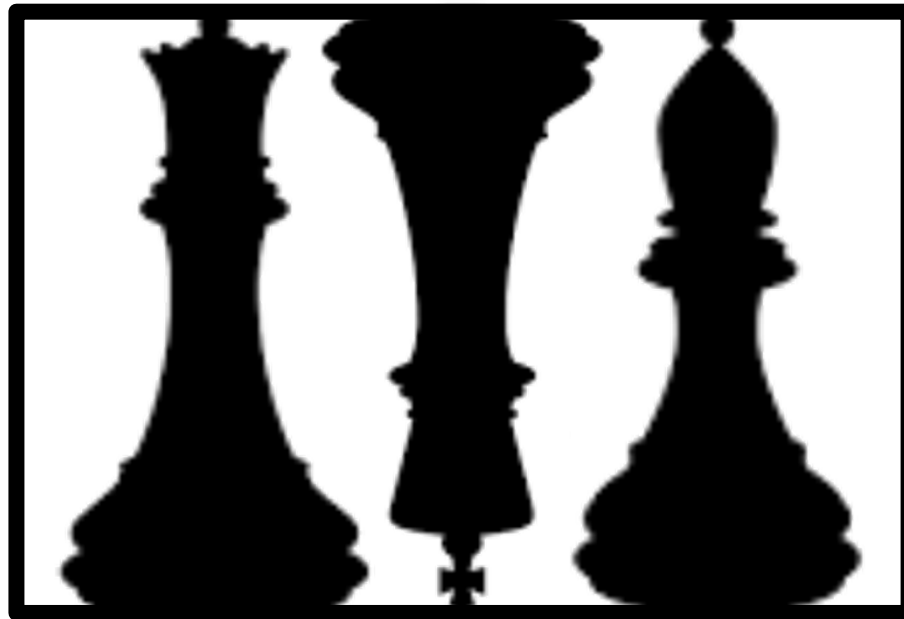
☐ Software

☐ Playlists

## Input: perception

☐ Visual

- Image
  - Pattern recognition: symmetry



How many objects?

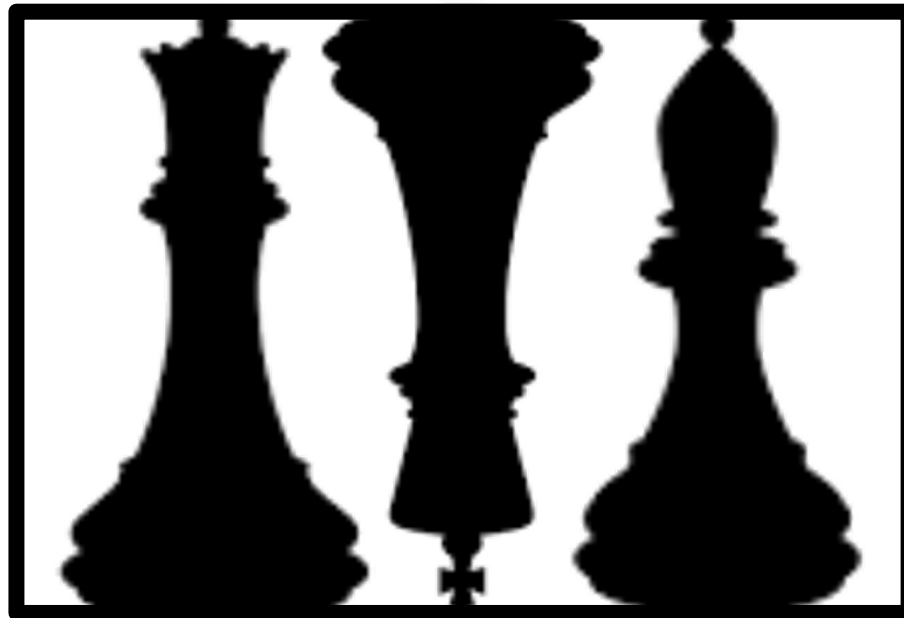
## Input: perception

☐ Visual

Symmetrical images are perceived collectively, in spite of distance

- Image

- Pattern recognition: symmetry



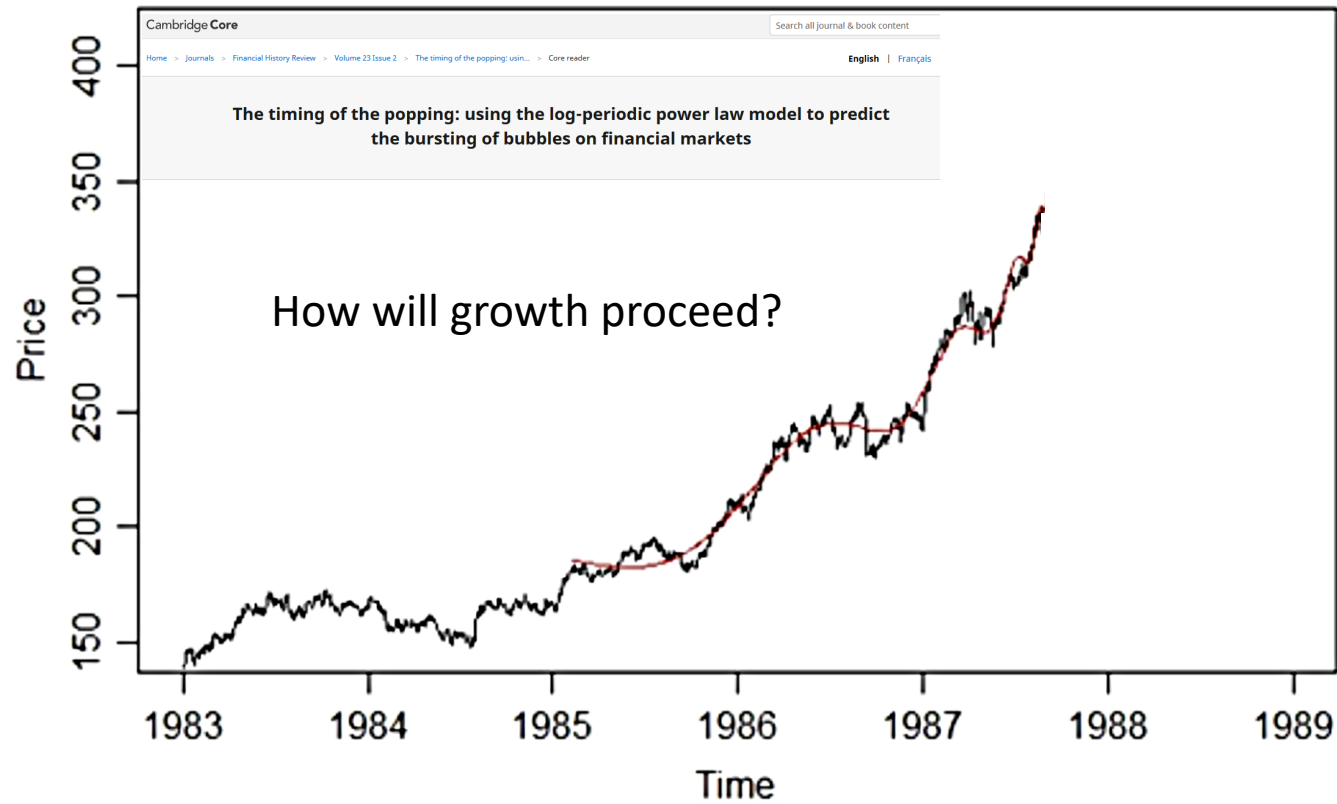
In the rectangle, we will most likely perceive three black symmetrical shapes against a white background (rather than four asymmetrical white objects against a black background)

## Input: perception

☐ Visual

- Image

- Pattern recognition: continuity

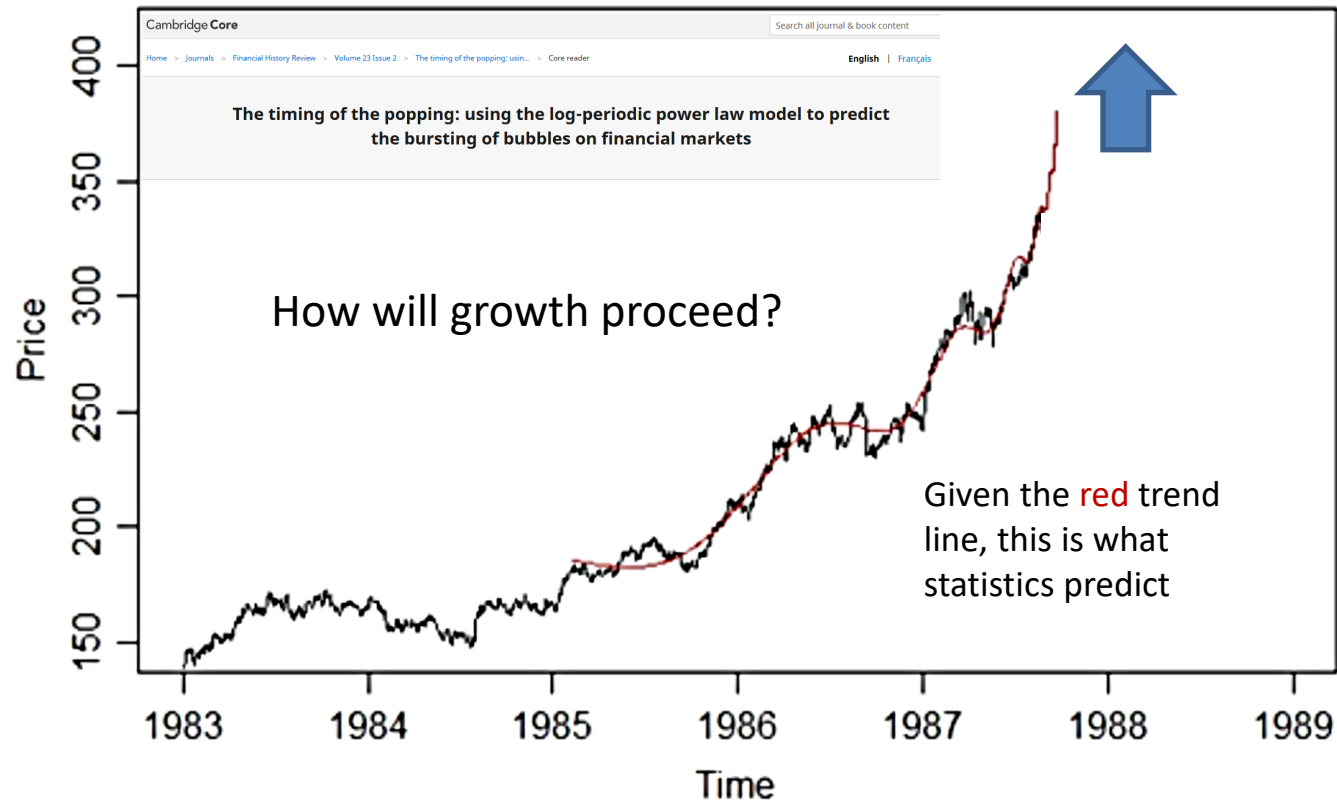


## Input: perception

☐ Visual

- Image

- Pattern recognition: continuity

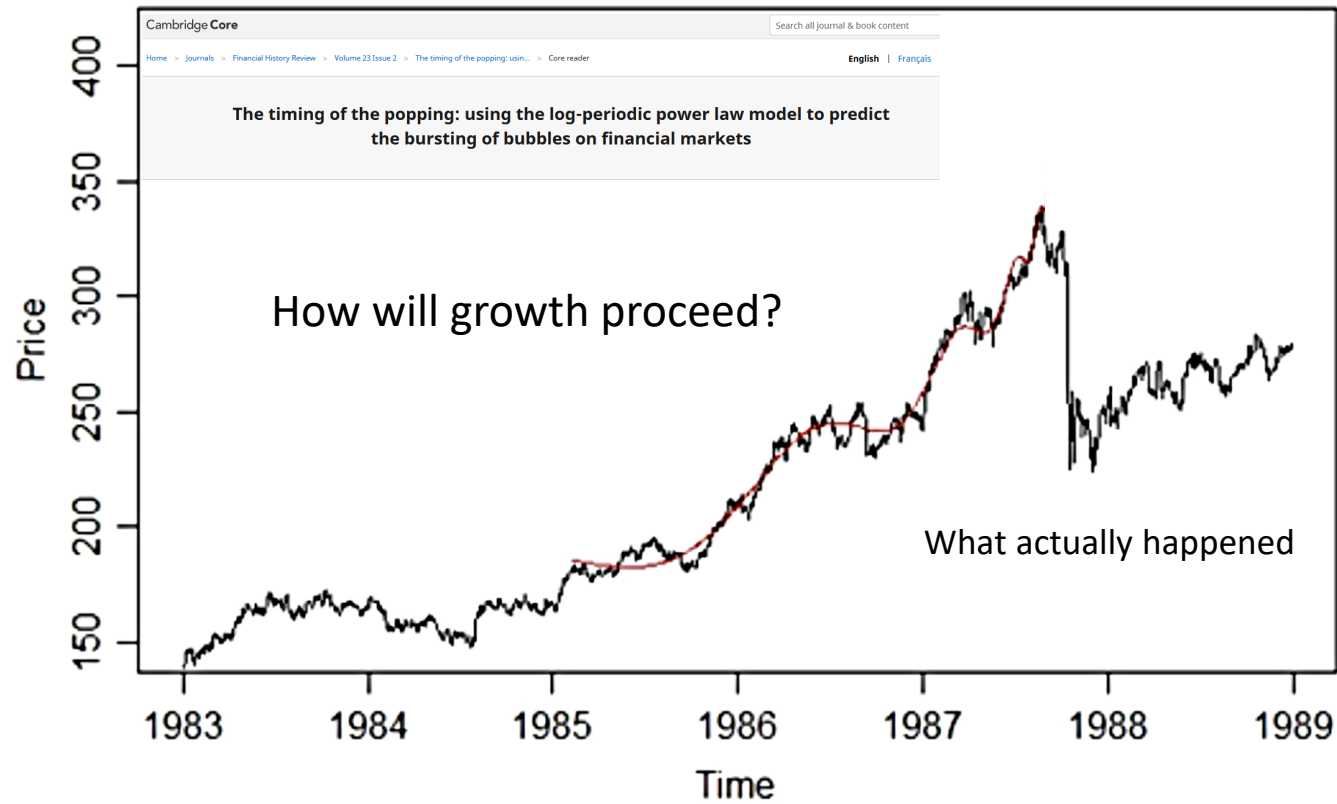


## Input: perception

□ Visual

- Image

- Pattern recognition: continuity



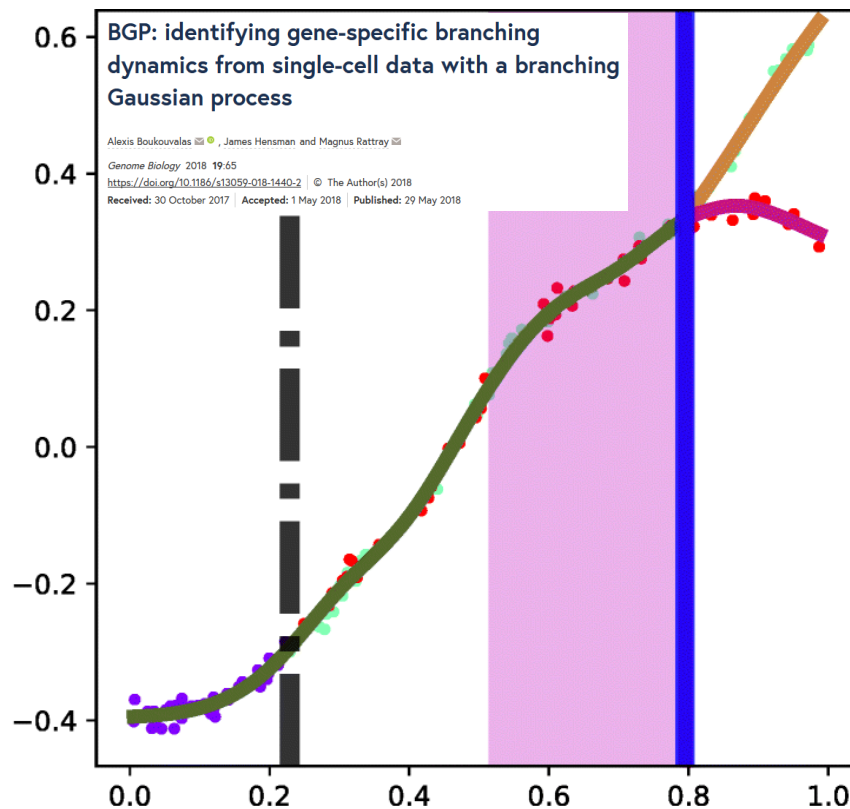
## Input: perception

☐ Visual

Lines are seen as following the smoothest path

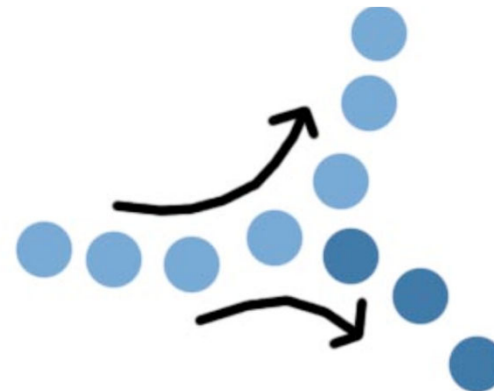
• Image

- Pattern recognition: continuity



(b) BGP Low noise

The top branch is seen as continuing the top segment of the line, rather than as an independent line



## Input: perception

☐ Visual

- Image
  - Pattern recognition: closure



How many objects?

## Input: perception

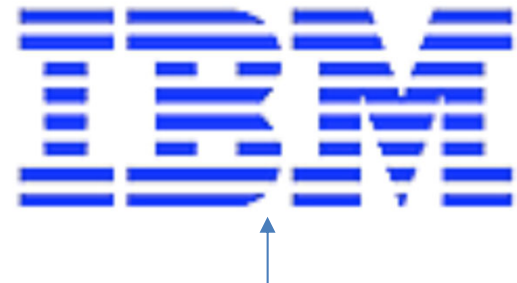
### ☐ Visual

- Image
  - Pattern recognition: closure

Objects grouped together are seen as a whole. We tend to ignore gaps and complete lines

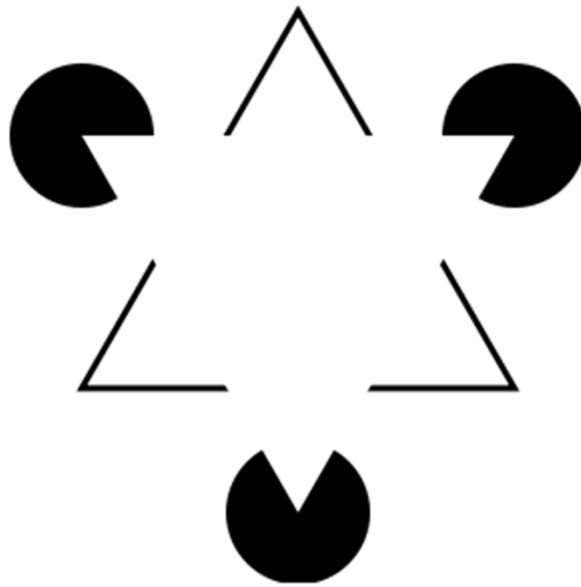


How many objects?



Blue lines on a white background.  
Or? A blue background with slits  
in the white paper?

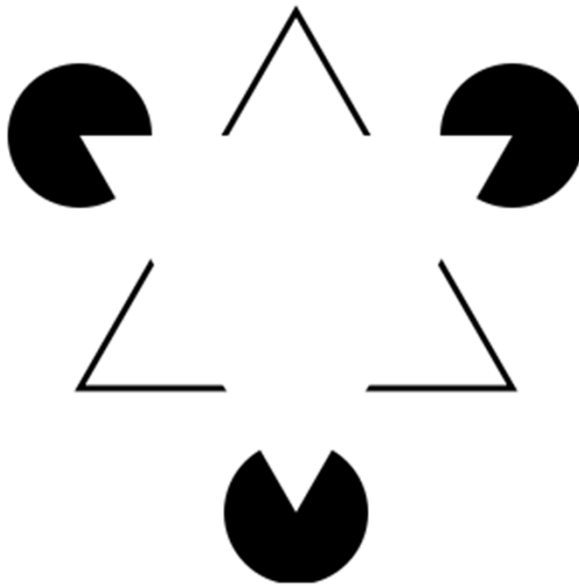
How many triangles?  
How many circles?  
How many stars?



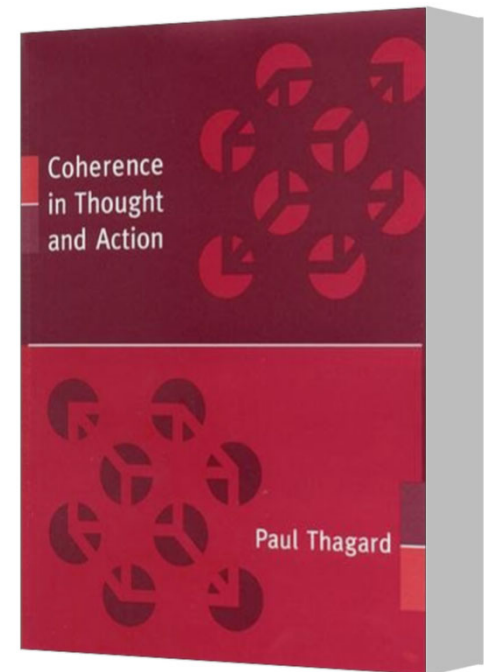
Kanizsa triangle

How many triangles?  
How many circles?  
How many stars?

There are no circles  
or triangles in this  
picture. But the  
brain “fills in the  
gaps” with familiar  
figures



Kanizsa triangle



## Deep Learning for visual recognition



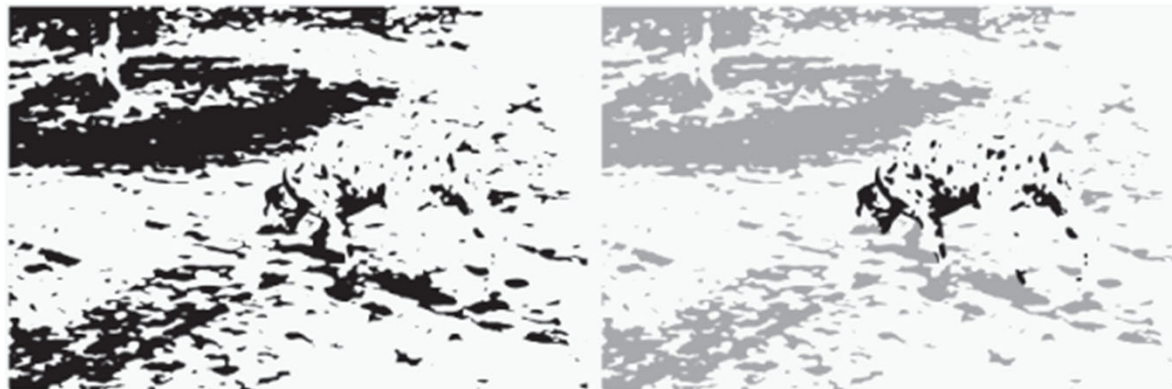
What should the system recognise?

## Deep Learning for visual recognition



What should the system recognise?

## Deep Learning for visual recognition



## Deep Learning for visual recognition



Yet, black dots are just  
unconnected data  
points. Conceptual  
interpretation makes it a  
Dalmatian dog





## Input: perception

☐ Visual

- Image

- Pattern recognition: closure

1; 2; 3; 4; 5; \_; 7; 8; 9;

Data imputation. What should be in the empty place?

## Input: perception

☐ Visual

- Image

- Pattern recognition: closure

1; 2; 3; 4; 5; \_; 7; 8; 9;

Data imputation. What should be in the empty place?

The most rational answer is: “I don’t know”



Lotto machine has 70 fair balls

Lotto host draws 7 balls

What is the chance you win the prize?

**The chance you win the prize is 1 in 1,198,774,720 but that's not important**

Next, you fill out 1 2 3 4 5 6 7

Your neighbour fills out 9 21 34 45 56 63 70

Who has the better chance of winning?

**No one. Finding new patterns in Big Data is a chance game**



The lotto host fills 7 balls with glue to make them heavier and fall through the hole

What is the chance the lotto host wins the prize? **100%**

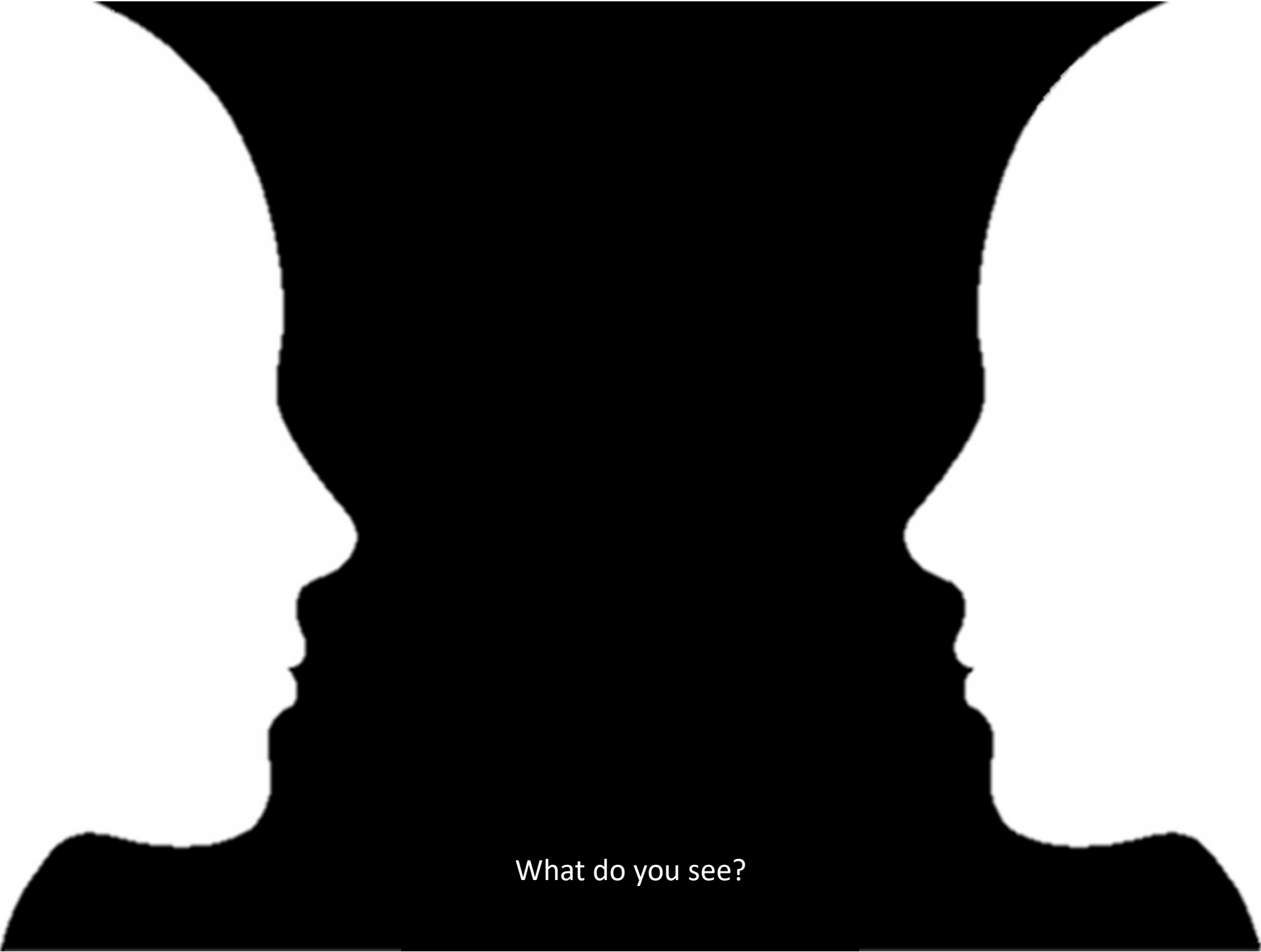
What is the chance you win the prize? **Same as before**

**Be careful with consulting GPT: "If the glued balls are guaranteed to be drawn, the host always wins and you have no chance." Not true, you coincidentally can pick the same numbers as the host**

**Chance depends on knowledge**

**The less you know the more probabilistic the problem space**

**So is innovation a deterministic or a probabilistic problem space?**

The image features a large, solid black shape that resembles a stylized hourglass or a central void. Within this black shape, there are two white, symmetrical profiles of human faces facing each other. The profiles are positioned such that their heads are at the top and their chins are at the bottom, creating a central white space where they appear to be looking at each other. The overall composition is balanced and minimalist.

What do you see?



Now, what do you see?  
(as well)

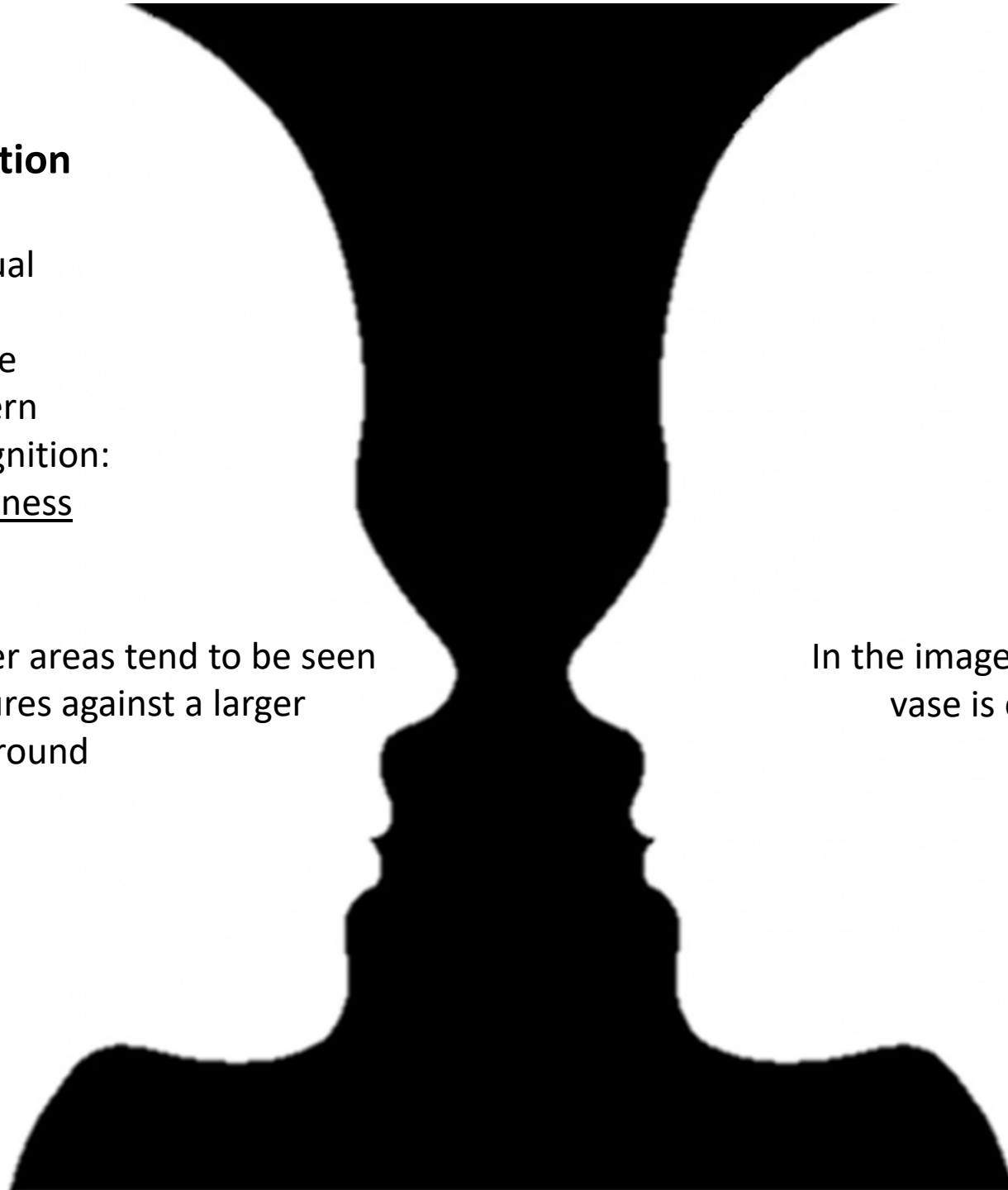
## Input: perception

☐ Visual

- Image
  - Pattern recognition:  
smallness

Smaller areas tend to be seen as figures against a larger background

In the image, “seeing” the vase is easier when it is smaller





What are these?



What are these?



What are these?



What are these?

They are letters



The letters spell the word TIE

## Input: perception

□ Visual

- Image

- Pattern recognition: surroundedness

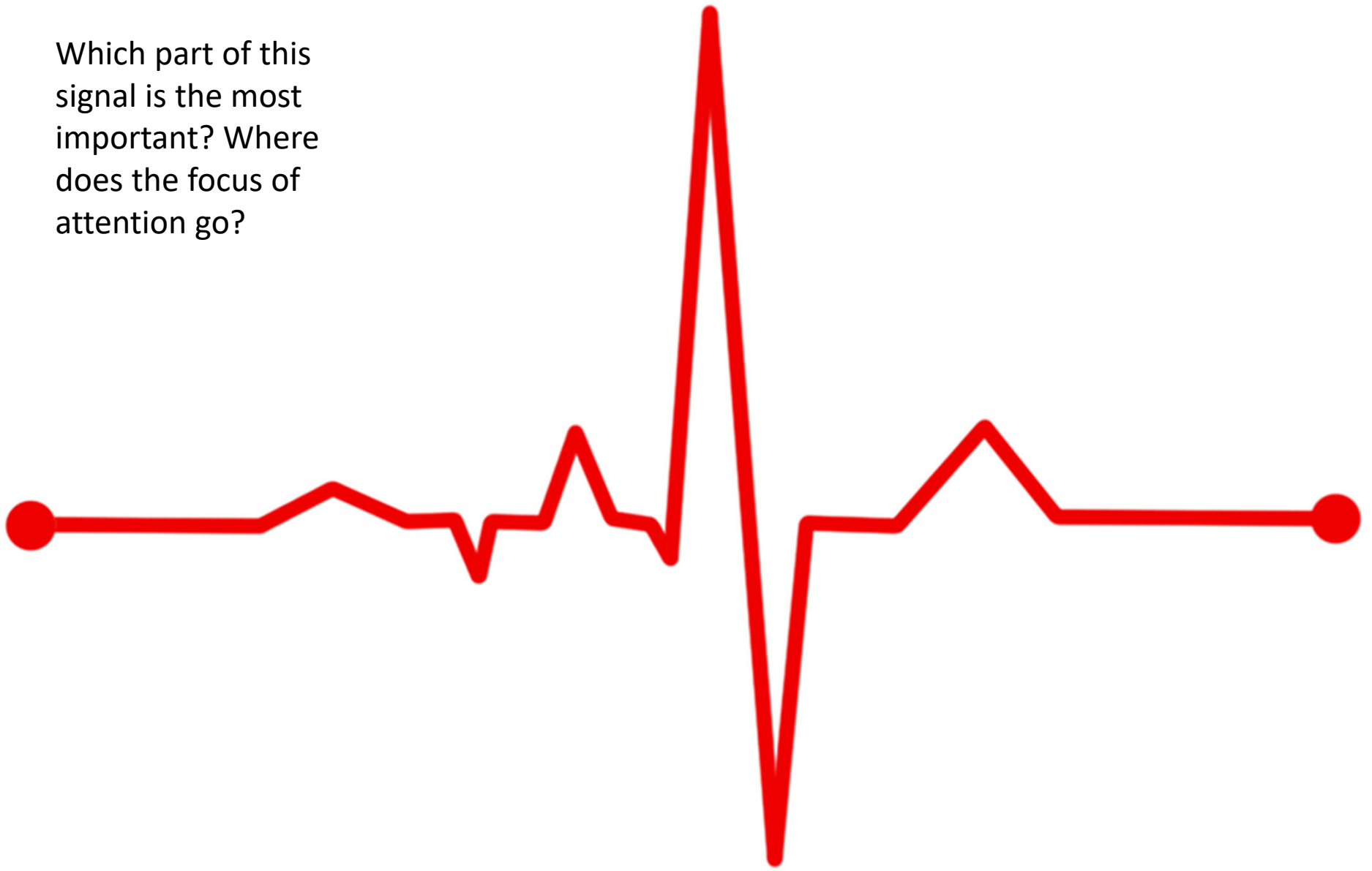
Areas which are surrounded by others tend to be seen  
as independent figures



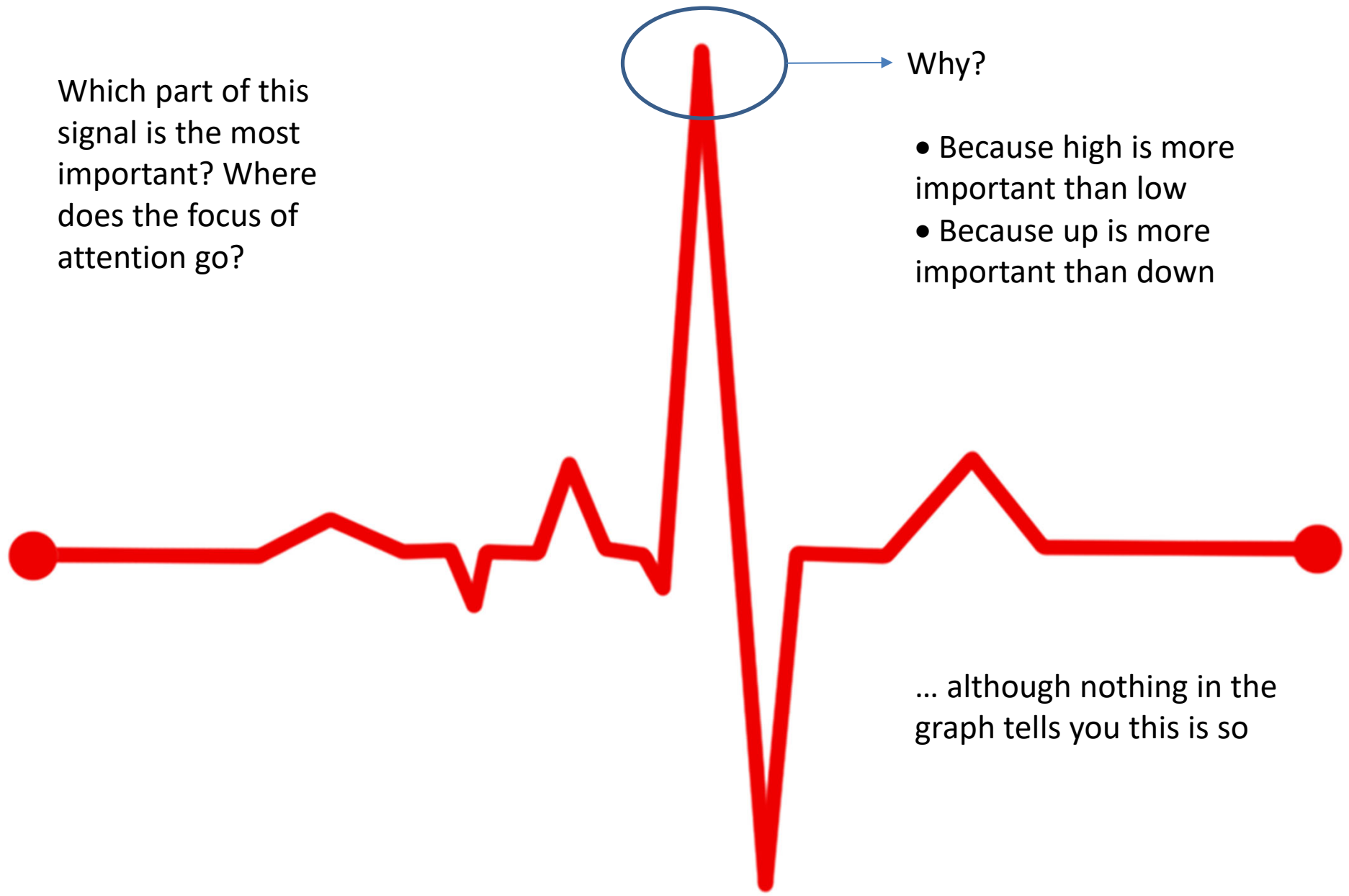
At first, most people will not see the word “TIE” in white letters  
against a black background



Which part of this signal is the most important? Where does the focus of attention go?



Which part of this signal is the most important? Where does the focus of attention go?



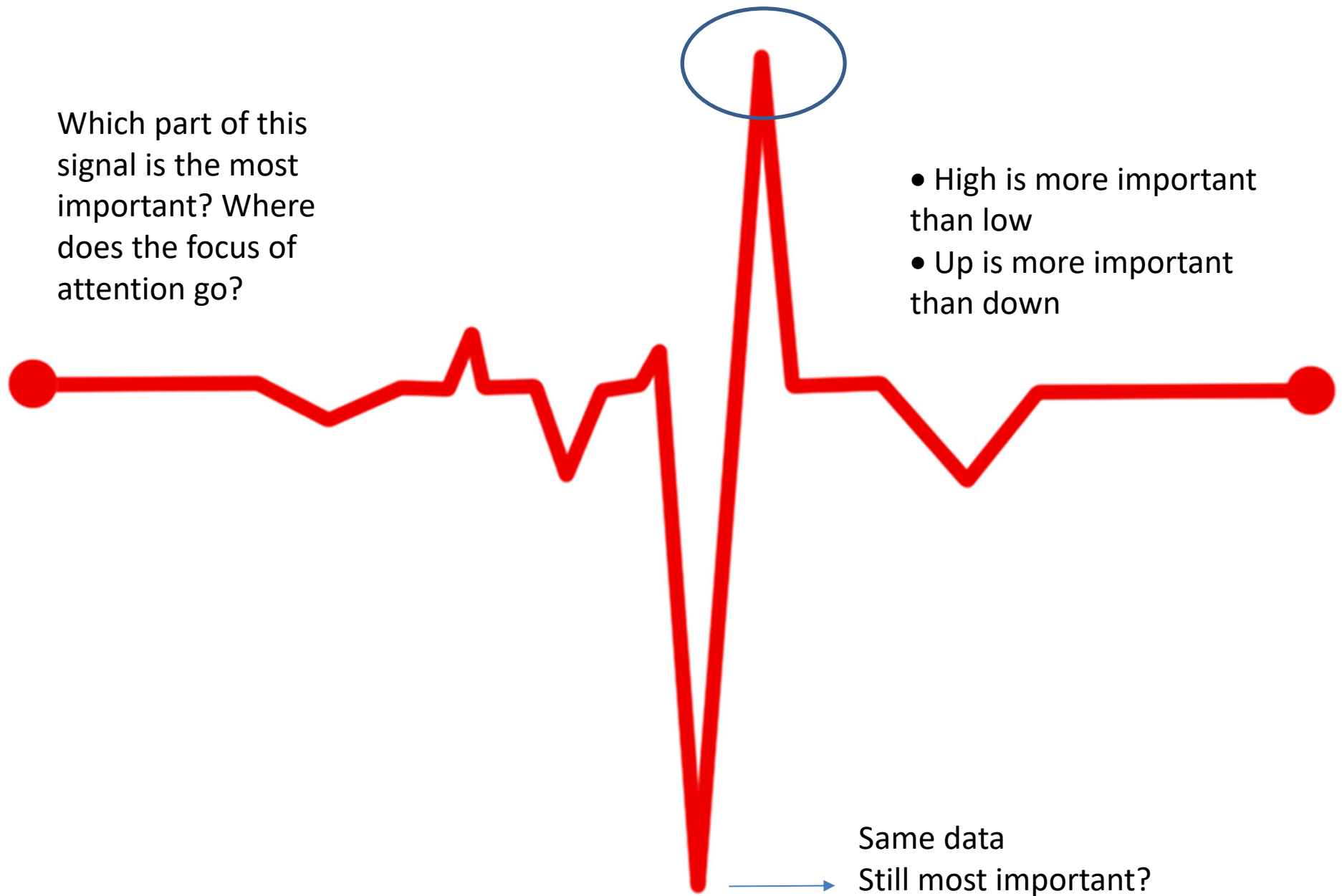
Why?

- Because high is more important than low
- Because up is more important than down

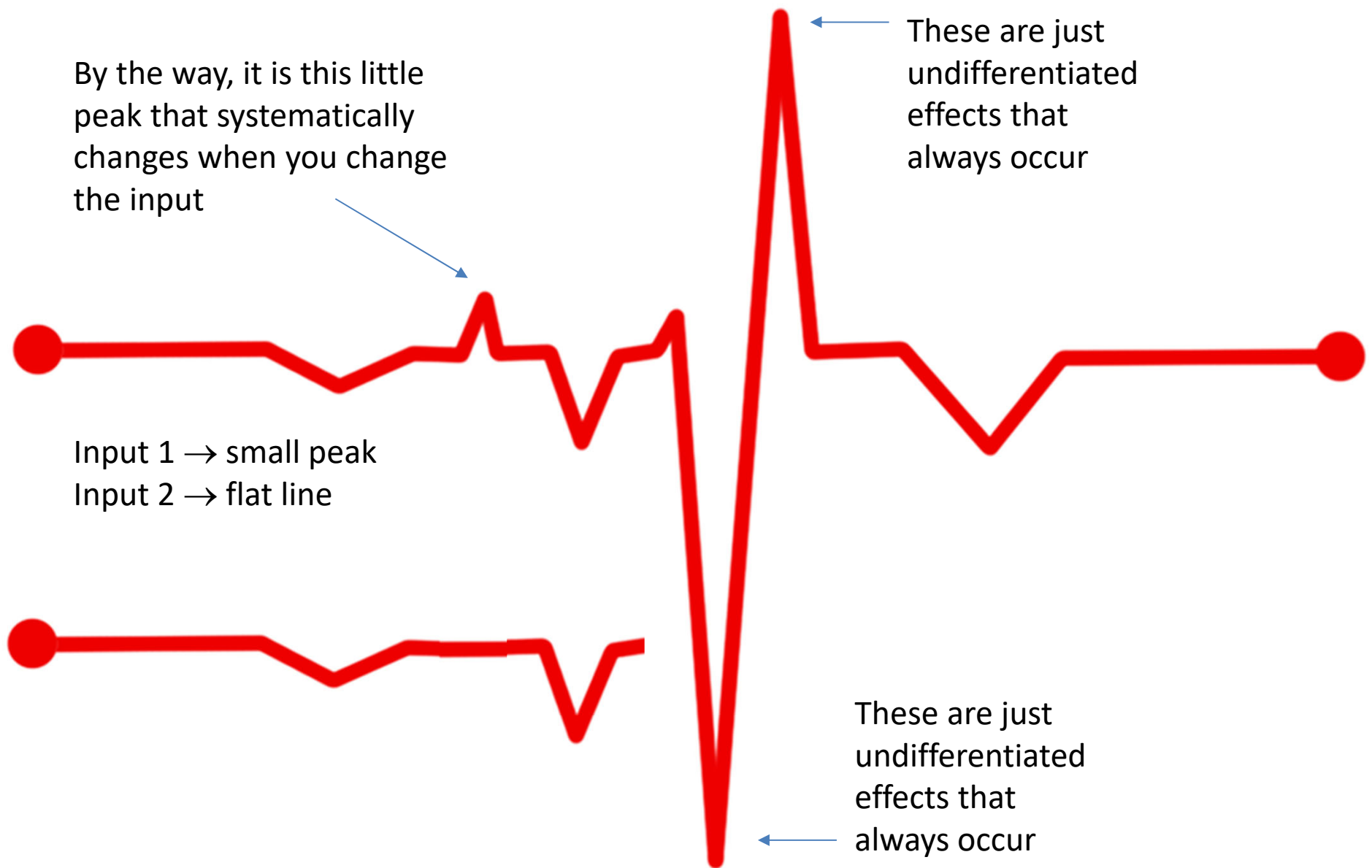
... although nothing in the graph tells you this is so

Which part of this signal is the most important? Where does the focus of attention go?

- High is more important than low
- Up is more important than down



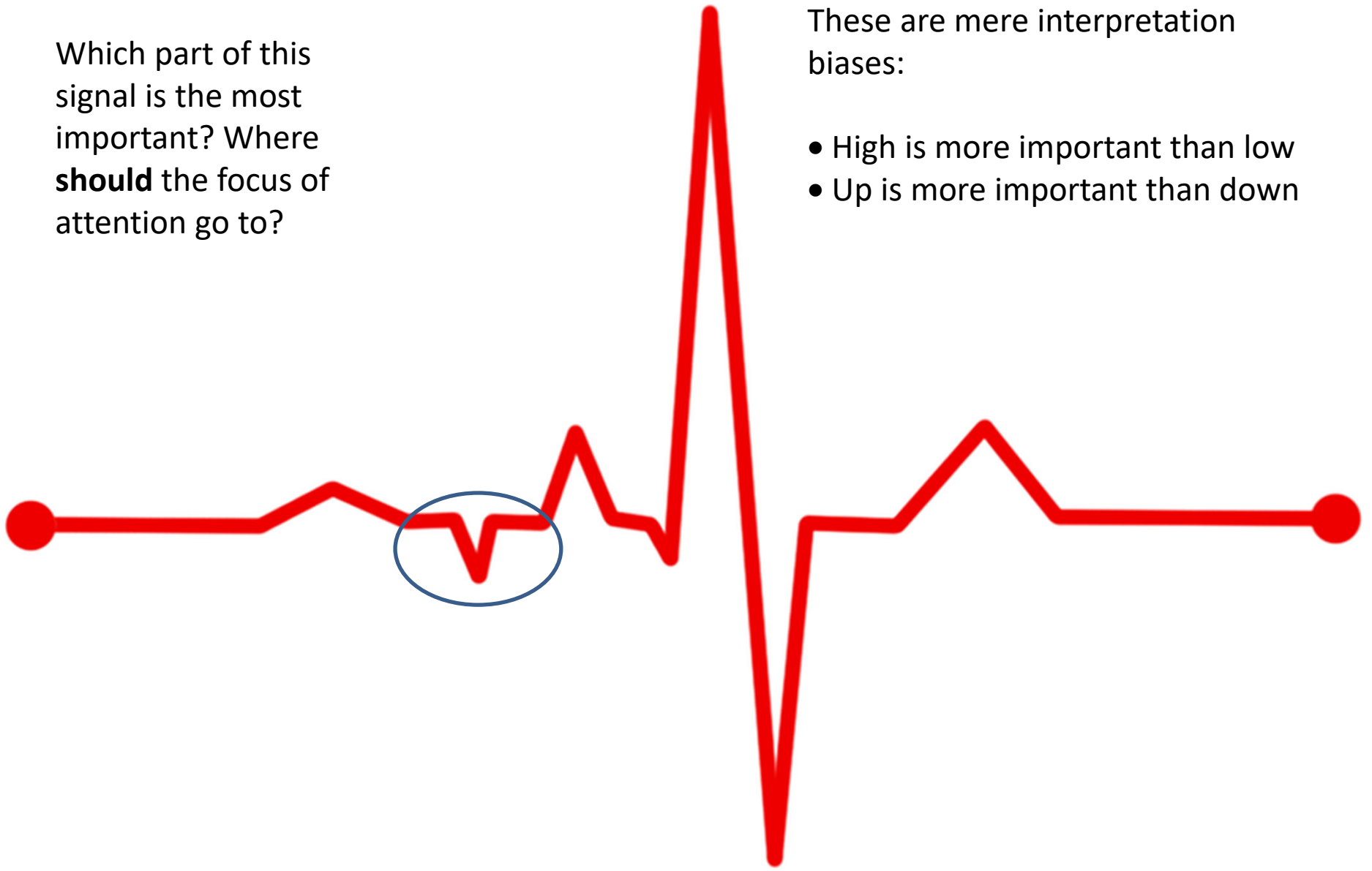
Same data  
Still most important?



Which part of this signal is the most important? Where **should** the focus of attention go to?

These are mere interpretation biases:

- High is more important than low
- Up is more important than down



## Input: perception

☐ Visual

- Image
  - Pattern recognition

### Consequences

Therefore, pattern recognition is **NO** property of the data (counting frequencies may do just as well)

To turn data into information, the following happens:

DATA —————> INFORMATION

# Input: perception

## ❑ Visual

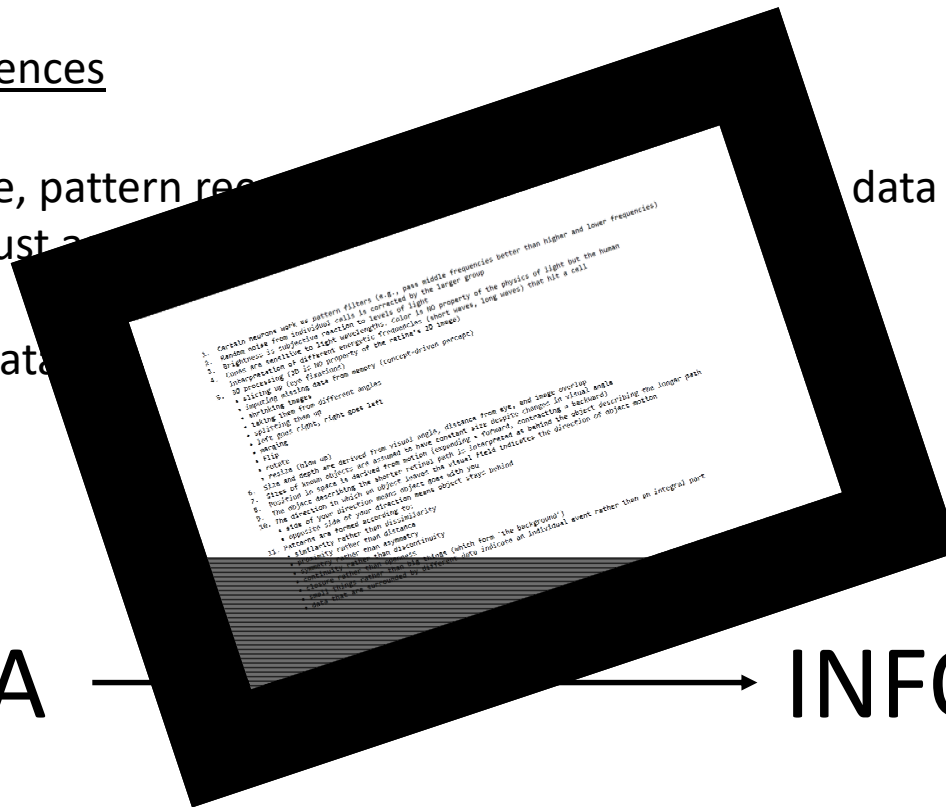
- Image
- Pattern recognition

## Consequences

Therefore, pattern recognition is a data processing task (counting frequencies)  
may do just as well as a human

To turn data

DATA



INFORMATION

# DATA

1. Certain neurons work as pattern filters (e.g., pass middle frequencies better than higher and lower frequencies)
2. Random noise from individual cells is corrected by the larger group
3. Brightness is subjective reaction to levels of light
4. Cones are sensitive to light wavelengths. Colour is NO property of the physics of light but the human interpretation of different energetic frequencies (short waves, long waves) that hit a cell
5. 3D processing. 3D is NO property of the retina's 2D image
  - slicing up (eye fixations)
  - imputing missing data from memory (concept-driven percept)
  - shrinking images
  - taking them from different angles
  - splitting them up
  - left goes right, right goes left
  - merging
  - flip
  - rotate
  - resize (blow up)
6. Size and depth are derived from visual angle, distance from eye, and image overlap
7. Sizes of known objects are assumed to have constant size despite changes in visual angle
8. Position in space is derived from motion (expanding = forward, contracting = backward)
9. The object causing the shorter retinal path is interpreted as behind the object causing the longer path
10. The direction in which an object leaves the visual field indicates the direction of object motion
  - the side of your direction means object goes with you
  - opposite side of your direction means object stays behind
11. Patterns are formed according to:
  - similarity rather than dissimilarity
  - proximity rather than distance
  - symmetry rather than asymmetry
  - continuity rather than discontinuity
  - closure rather than openness
  - small things rather than big things (which form 'the background')
  - data that are surrounded by different data indicate an individual event rather than an integral part

# INFORMATION

## Human-generated Summary



We casually speak about data as if they are lying around like rocks and sand and we habitually 'extract' information from the data as if chemical content (polyphenol) seeps out of infused green tea leaves

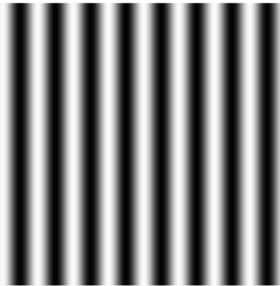
However, information is not neutral. The eye filters, selects, and changes data so to make an interpretation of the world about us. Colour is not a property of light but of the eye's receptors. 3D vision results from manipulations by the brain to create a depth illusion. The brain uses heuristics such as similarity, continuity, and proximity (so-called 'Gestalts') to make sense of information that passes the eye. Patterns are not the result of recognition but rather of human construction

**ALL** computer analysis depends on human analysis!!

## Visual Stimuli and Cell Responses

### Learn about visual stimuli

Drifting Gratings



Static Gratings



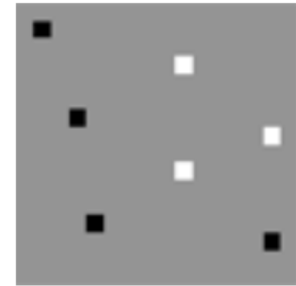
Natural Scenes



Natural Movies

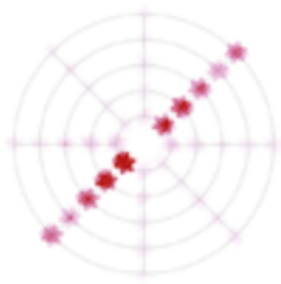


Locally Sparse Noise

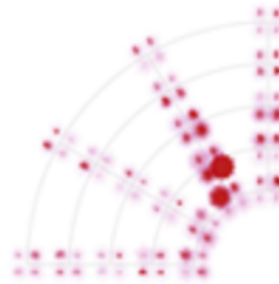


### Browse cell responses

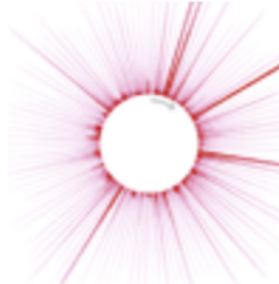
Drifting Gratings



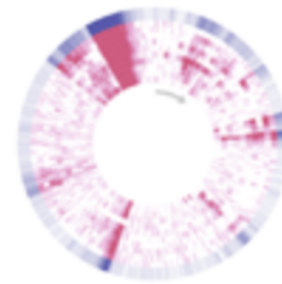
Static Gratings



Natural Scenes



Natural Movies



Locally Sparse Noise

